



A survey of fault prediction and location methods in electrical energy distribution networks

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ABSTRACT

One of the main factors that disrupt reliability and stop energy provision is the fault occurrence in distribution networks. Thus, accurate and fast fault prediction and location in distribution networks are essential for increasing reliability, fast restoration, optimal electrical energy consumption, and customer satisfaction. This study reviews and investigates fault prediction and fault location topics. To this end, the existing methods and views in the context of fault prediction are reviewed first; then, fault location is investigated. This paper investigates various methods, their advantages, disadvantages, technical reports, and patents in conventional distribution networks, smart-grids, and micro-grids. Comparison of this study with other surveys indicates that it is more comprehensive and despite others covers fault prediction. In addition, it includes an up to date review of the methods for distance measurement and fault location considering different network types (AC/DC), presence of DG, communication and automation standards, synchronous and unsynchronous measurement, magnetic measurement, and state estimation-based fault location methods.

1. Introduction

One of the essential tasks of the power system operators is the fault location. The fault might result in problems like network device damages, service interruption, the network instability, hence, reducing the network reliability. Accordingly, financial losses are imposed on the customers and electricity companies. The traditional fault location methods in feeders of distribution networks are not efficient in particular when the geographical distribution of the network is vast. Covering a vast area is expensive both in terms of the required manpower and devices and also takes time. Therefore, predicting and locating fault automatically and quickly are important in distribution networks. Automatic fault prediction and location have advantages like saving time, human resources, enhancing system readiness to maintain power, modifying the future schedule, and improving economic factors. These factors increase customers' satisfaction and improve the reliability indices of the system [1–3].

Fault prediction is the analysis and mining of historical data for

predicting the absence or presence of a fault in the power system. Pattern recognition methods and machine learning algorithms have become important for fault prediction in power systems [2].

According to the literature, various methods have been presented to measure fault distance in transmission networks. However, the presented methods cannot be used to locate faults in the distribution networks due to structural differences in the distribution network compared to the transmission network. Thus, impedance methods, differential equations, traveling waves, fault indicators and magnetic sensors, protection coordination and current analysis, state estimator, and artificial intelligence methods have been presented to solve the fault location problem in distribution networks. However, these methods have some problems. The main problem of employing impedance methods in distribution networks is the multiple response problem. The differential methods have some disadvantages, such as requiring connection links with high bandwidth, high sampling rate, fast and accurate data synchronization. The traveling wave methods have a complicated structure and require a high sampling rate. Using magnetic

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sensors and fault indicators is uneconomical because of the distribution network's size. The main disadvantage in the methods of protection coordination and current analysis is the interference in the detection of the main section of the fault. This problem occurs when the fault resistance is high and affects the current domain. As a result, the performance of the relay is affected. The need for high sampling rates and accurate line parameters, and also complexity are disadvantages of state estimation methods. Finally, requiring accurate and large data banks that should be updated when a small change is applied to the network is the disadvantage of intelligent methods [3,4].

In this part, presented studies in the context of fault location and prediction in distribution networks, are compared and investigated. Table 1 shows the results of this comparison. In this table, comparisons are made based on the investigated methods, network type, smart grid, state estimation, presence of DG, fault type, fault section estimation, fault distance determination, fault prediction, load estimation, line parameters estimation, line model, unbalanced system, load model, communication and automation standards, magnetic measurement, and synchronous/unsynchronous measurement. To the best of our knowledge, this review paper is the only work which reviews the state-of-art in fault prediction. In addition, it also gives an up to date comprehensive overview of different fault location methods.

The rest of the paper is organized as follows. In Section 2, firstly, various faults are introduced for power systems. Secondly, unsystematic and systematic fault prediction using various methods, including SVM, neural network, etc. are analyzed. Fault distance determination and fault section estimation methods in the conventional distribution network, smart grid, and microgrid are evaluated, and operations, advances, pros, and cons of the proposed methods are studied in Section 3. The future research is presented in Section 4. Finally, the paper is concluded in Section 5.

2. Fault prediction methods in electrical distribution networks

Faults may occur in distribution networks due to lightning, insulation defect, sabotage, tree branches, and animals, which result in a short circuit. Many faults are naturally transient, which are resolved without losing supply or with minimum interruption time. Persistent faults result in longer interruptions, and they are resolved after fault detection and location [1].

Faults in three-phase AC systems are classified as asymmetric and

symmetric faults based on the types shown in Fig. 1. Examples of asymmetric faults include a single phase to ground short circuit fault (LG), two-phase short circuit fault (LL), and two-phase short circuit fault to ground (LL-G). On the other hand, symmetric faults also include three-phase faults such as three-phase short circuit (LLL) and three-phase short circuit to ground (LLL-G).

There are two types of faults in DC networks: pole to pole (PP) and pole to the ground (PG). PP faults occur when the conductors are connected directly. Thus, the resistance of PP faults is low, and it is known that low-resistance faults are more dangerous. PG faults occur when one of the conductors or both are connected to the ground. Thus, the resistance of PG faults is high [14].

Faults occur in energy systems and cause hazardous transients, equipment failure, and power outages that decrease system reliability and result in customer dissatisfaction. To move from the reactive and inefficient maintenance approaches to a more proactive maintenance strategy, fault prediction plays an important role. If we detect early symptoms before a fault manifest, we can predict fault in the energy systems components, and start the remedial actions early in the process and prevent faults and failure. For example, we can change or repair the components which show symptoms (a cable with a spark in it, a transformer with partial discharge, etc.). These devices malfunction in such a way that in case of any abrupt transitions (thunder, overload, etc.), they may fail and cause the whole system to shut down. The ability to care for systems and components in a healthy condition by predicting what happens in the future is known as predictive maintenance [15,16]. In electrical distribution networks, faults can be predicted using machine learning tools and model training designed by employing real and/or simulated data. This approach allows the system to prevent the fault from happening by predicting it and increasing the system's reliability. To this end, the real data of the studied system must be categorized. These data may include the recorded weather condition from weather stations and the failure records with geographical coordinates or even periodic recorded voltage and current of the energy system component. Therefore, two general cases may exist based on the data type: (i) predicting fault in an electrical network using weather condition and/or characteristic data of systems components, (ii) predicting fault in the equipment of the network such as transformer, cable, etc. using periodic recorded voltage and current, load value, etc. In the second case, there is an assumption of having labeled data. On the other hand, fault can be divided into two general cases of systematic and unsystematic faults.

Table 1
Comparison of this paper and others presented in the context of fault location and prediction.

References	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	This paper
DG	–	✓	✓	✓	✓	–	✓	✓	✓	✓
Smart grid	–	Automation	✓	✓	✓	Automation	–	✓	✓	✓
Fault Prediction	–	–	–	–	–	–	–	–	–	✓
Fault Distance	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Section Estimation	✓	✓	✓	✓	–	✓	✓	–	–	✓
Network Type	AC	AC	AC	DC	AC	AC	AC	DC	DC	AC-DC
Fault Type	All	All	All	PP-PG	All	All	All	PP-PG	PP-PG	All
Methods	ML	IB-TW-ML	IB-TW-ML	TW-ML	TW	IB-TW-ML	TW-IB-ML	IB-TW-SCS	TW-IB-SCS	All
State Estimation	–	–	–	–	–	–	–	–	–	✓
Load estimation	–	✓	–	–	–	–	–	–	–	✓
Line Parameters estimation	–	–	–	–	✓	✓	–	–	✓	✓
Line Model	✓	✓	–	–	–	–	✓	–	✓	✓
Unbalanced system	–	✓	✓	–	–	–	✓	–	–	✓
Load Model	–	✓	–	–	–	–	✓	–	–	✓
Communication and automation standards	–	✓	–	✓	✓	✓	✓	✓	✓	✓
Magnetic measurement	–	–	–	–	–	✓	–	–	–	✓
Synchronous and Unsynchronous measurement	✓	✓	✓	✓	✓	–	✓	✓	✓	✓

TW = Traveling Waves.

IB = Impedance Base.

ML = Machine Learning (Pattern Recognition, Artificial Intelligent, Knowledge-based).

PP = Pole-to-Pole.

PG = Pole-to-ground.

SCS = Sensors and Communication Systems.

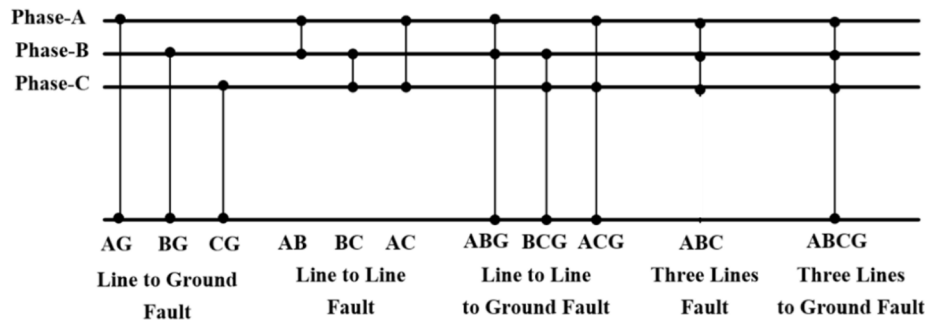


Fig. 1. Faults in power system [7].

The systematic faults are those that can be predicted using recorded data of the network by following the right patterns. Faults in transformers, power insulators, transmission cables belong to the systematic faults group. Unsystematic faults are related to weather conditions and human abruptions that cannot be predicted. Lightning, external object interference, and human-made faults are common examples of unsystematic faults. The fault prediction could be made by training a prediction model using sample data. It is run continuously on voltage and current data and detects the disturbances. According to defined indices, the disturbances which can cause a fault in the future are distinguished and introduces as a priority item for predictive maintenance [17–19]. In the past decades, various artificial intelligence methods have been investigated for fault prediction in power systems. For example, expert systems [20,21], Bayesian Network [22,23], rough sets [24,25], Petri nets [26,27], neural networks [28,29] etc., can be mentioned. Besides, various studies have focused on relay protection and performance of electrical devices [30,31]. System failure detection is to be optimized through testing failure assumptions. However, these processes have problems dealing with power system faults. Incorrect operation of relay protection and electrical elements has adverse impacts on the results. Studies on data-based methods have become more common in recent years. Recently, artificial neural networks (ANNs) have been widely used for fault prediction. The fault prediction model accuracy has been improved using an optimized ANN based on a multi-level genetic algorithm [32]. Recurrent neural networks (RNNs) can extract the hidden correlation of the big data. Also, they have shown sufficient capability to operate during faults [33]. However, the main RNN suffers from the vanishing gradient problem, because the understanding of the next nodes from the previous nodes decreases. Long Short-Term Memory (LSTM) networks outperform conventional RNNs in terms of feature extraction in a longer time interval. For example, the authors of [34] have presented a fault detection and prediction method based on the LSTM network. This method performs well in complex operations, hybrid faults, and strong noises. In general, LSTM is an improved RNN that is more compatible with long time series. However, studies on fault prediction presented based on data using LSTM are still in their infancy. Classification is one of the main steps of the fault prediction process. SVM is a discriminant classifier defined by a hyperplane [35]. Data mining based fault prediction method has been presented for distribution networks [36]. First, the effective factors (operational factors, device-related factors, and external factors) are analyzed. Next, the resources are investigated, and the obtained data are pre-processed. After that, the Relief F algorithm is used as a feature selection algorithm to effectively select the model input vectors. Then, the faults are classified based on fault frequency and fault range. Finally, SVM is used with PSO to construct the prediction model. In the technical report IEEE PES-TR73 [37], the effects of different equipment failures on the waveform and RMS graphs of voltage and current. This report uses different research works and voltage and current data extracted from the installed CTs and PTs at the beginning of the feeder. This information can be used to predict various faults. This reference has classified various equipment failures as follows:

- Cables (Incipient Failures of Cables, Failures of Cable Joints, Failure of Cable Terminations, Experimental Results on Cable Insulation Failure)
- Overhead Lines
- Transformers (Failures of Tap Changers, Failures of Transformer Bushing)
- Switches (Failure of Line Switch, Arcing of Capacitor Bank Switch, Restrikes during Capacitor De-energization)
- Capacitors (Capacitor Failure Caused by Misoperation of Controller, Unsuccessful Synchronous Closing Control, Capacitor Energization Triggering Resonance, Other types of failures in a capacitor bank)
- Lightning and Surge Arresters (Failure of Lightning Arresters, Failure of Surge Arresters)
- Potential Transformers (No corresponding response on the current channels, Only phase-A voltage is affected, Perturbation at voltage peak, Notching at zero crossings)

In the following two types of unsystematic and systematic fault prediction are reviewed.

2.1. Unsystematic fault prediction

In the technical report by EPRI¹ [38], the power quality disturbances have been used in distribution lines to present a number of criteria using which it has performed statistical and waveform analyses on transient faults such as lightning strike, and collision of animals and trees. For example, Fig. 2 shows the waveforms of three-phase voltage and current due to a lightning strike. This waveform has been obtained from power quality monitors in a 161-kW system.

In [39], the failure data obtained from a distribution network in Beijing is used for short-term prediction of failure. This data is shown in Fig. 3. First, the effect of weather conditions on the number of faults is studied to develop an SVM classifier. The factor with the maximum impact on fault occurrence is specified, and the factor with minimum impact is eliminated through analysis. Then, the classification prediction model of the distribution network (DN) short-term failure frequency (weekly) is developed to predict the number of DN faults. The accuracy of this method is reported as 70%.

In [40], the effect of various weather conditions on device failure (underground) in MV networks has been studied. Also, a prediction model has been presented using the basket analysis. First, the correlation between weather and failure is evaluated statistically. Then, several key weather factors of one day are classified to “high, medium and low” for big data analysis and develop a prediction model. This model predicts risk in various areas in the day ahead. In [41], a fault risk prevention model in DN based on the hidden Markov chain has been presented. This method combines various risk development conditions in a DN, like fault prevention, event, and control, to develop a state machine model. The

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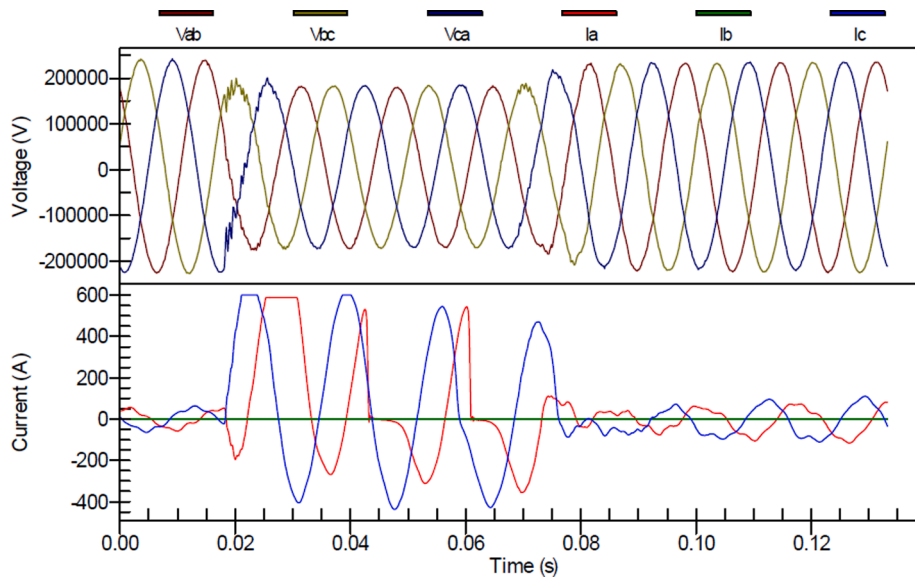


Fig. 2. Three-phase voltage and current waveforms of a lightning event captured at a 161-kV system [38].

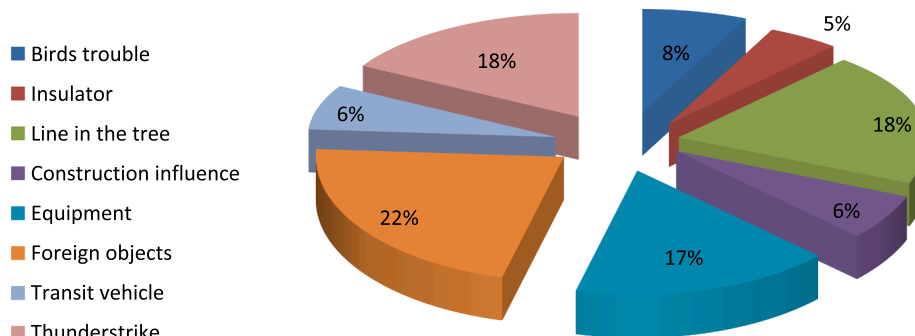


Fig. 3. Failure causes in a DN in Beijing [39].

state sequence of the fault events can be observed by the hidden Markov chain, and the hidden fault cause is excavated. The accuracy of this method is 82.8%.

2.2. Systematic fault prediction

2.2.1. Line trip fault prediction

One of the most common faults that occur in power systems is the line trip fault, which has been studied in [42,43]. If the recloser does not operate successfully, an outage occurs, and financial losses are imposed. In [2], a method has been presented for line trip fault prediction in power systems using the LSTM network to extract time features and SVM for classification. Current, voltage and active power measurements before trip faults or during normal operation are selected as input. The parameters can be learned offline and updated online. The accuracy of different methods (BPNN², SAEs³, RNNs, SVM, and LSTM) in this area are shown in Fig. 4 [2].

In [44], a probabilistic sequence classification method is presented to predict early faults in distribution grids. In this work long-short term memory (LSTM) neural network is used. Typically, the machine learning approach and historical electrical network data are used to predict the risk of failure for the distribution network ingredients. This will give us the opportunity of replacing the improper equipment of the network

which is more likely to fail in the next severe network conditions (bad weather, lightning, fog, part of the network, etc.). Many works have been published which have presented a model to predict early faults using high-resolution data [45–47]. Despite recent works, in [44] low-resolution data are applied for early faults in the network. The data are gathered from pre-existing data collection devices and fed to LSTM neural network model. This significantly reduces the cost of predictive maintenance. In this work, a model is presented that can predict the state of different areas of the network for providing a general early warning system. This system can detect and predict the fault in the network before it happens which gives an opportunity to prevent the network from shutting down. An important point in the fault prediction problem is the predictability of fault. For instance, network faults such as object falling on the cable and human disruptions cannot be predicted using data-driven methods because these do not cause any specific signature in the network signals before their happening. Faults that occur due to aging and corrosion in the network which affect the network parameters such as voltage, current and frequency can be predicted. In the following LSTM neural network method; which is a variant of recurrent neural networks (RNNs); as a leading advanced approach for modeling sequential data is summarized.

• LSTM neural network

LSTM is an artificial recurrent neural network which is used in the deep learning field. In spite of conventional neural networks, it contains feedback connections in its structure. This feature allows it to be used in

² Back Propagation Neural Networks

³ Stacked Autoencoders

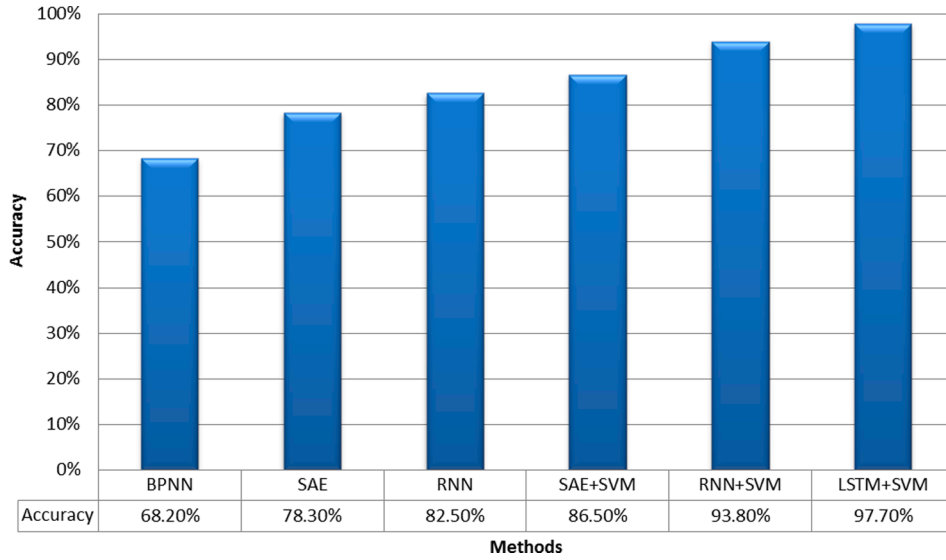


Fig. 4. Accuracy of various trip fault prediction methods [2].

sequential processes such as fault prediction in the distribution network using time series data. A simple computational node of an RNN is depicted in Fig. 5. It contains input, output, computation block and internal states. A feedback structure of its states exists in RNN. The LSTM module consists of four gates: forget, input, update and output gates and a cell state. The mathematical description of forget gate's output is as follows:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (1)$$

It is important to decide what information is going to set in cell state c_t . The update and input gates which have the output values in the boundary of $[-1 \ 1]$ and $[0 \ 1]$ are described as follows:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (2)$$

$$u_t = \tanh(W_u \cdot [h_{t-1}, x_t] + b_u) \quad (3)$$

To move the only relevant part to the next module the following equation is needed:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (4)$$

The two hidden states can be calculated from the two following equations:

$$c_t = c_{t-1} \odot f_t + i_t \odot u_t \quad (5)$$

$$h_t = o_t \odot \tanh(c_t) \quad (6)$$

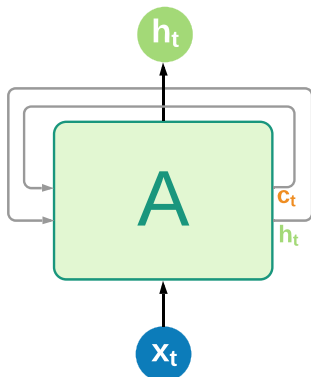


Fig. 5. RNN computational node [44].

• Modelling

In order to predict the fault in the underground cables in the distribution network, low-resolution data of voltage, current and power have been used. Due to the reduced accuracy of fault prediction, the measured raw data cannot be used and instead the fault prediction is done by defining state goals for each parameter. State goals for voltage (V_g), current (I_g) and power (P_g) are defined as follows:

$$V_g = \begin{cases} 1 - \frac{|V - \bar{V}|}{V_{max} - V_{min}}, & \text{if } V_{max} \neq V_{min} \\ 0, & \text{if } 1 - \frac{|V - \bar{V}|}{V_{max} - V_{min}} < 0 \end{cases} \quad (7)$$

$$I_g = \begin{cases} 1 - \frac{|I - \bar{I}|}{I_{max} - I_{min}}, & \text{if } I_{max} \neq I_{min} \\ 0, & \text{if } 1 - \frac{|I - \bar{I}|}{I_{max} - I_{min}} < 0 \end{cases} \quad (8)$$

$$P_g = \begin{cases} 1 - \frac{|P - \bar{P}|}{P_{max} - P_{min}}, & \text{if } P_{max} \neq P_{min} \\ 0, & \text{if } 1 - \frac{|P - \bar{P}|}{P_{max} - P_{min}} < 0 \end{cases} \quad (9)$$

One of the most common difficulties with predicting is the lack of real data. Hence, in [44], virtual data is generated using Gaussian distribution. The Gaussian distribution with the mean value μ and variance σ^2 is as follows:

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (10)$$

The generated sampling data are different from the real one as depicted in Fig. 6.

The results of this work show that it has good accuracy to predict fault in the network. The accuracy percentage of the model for the test data set for three networks' operation modes of normal, early warning, and critical are 0.932, 0.842, and 0.921.

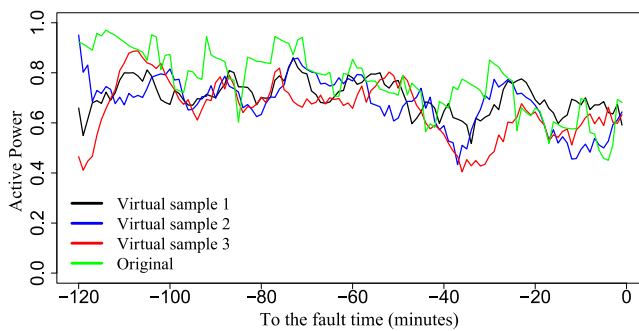


Fig. 6. Comparison of generated data and the real data for a specific faulty situation [44].

2.2.2. Voltage sag fault prediction

Voltage sag is perturbations of the voltage supply at any point of the power system, which reduces the amplitude of the voltage amplitude and voltage restoration after a short time. Voltage sag is the result of fault occurrence or sudden load increase in the power system. Their characteristics depend on fault location, fault characteristics, and protection systems. Since the faults occur at different times and locations, voltage sag has a probabilistic nature [48]. In [49,50], the fault position method and Monte-Carlo simulation method have been compared. It is concluded that both methods can generate similar results. One disadvantage of the fault position method is that fault resistance is usually considered to be constant and equal to zero. Therefore, low accuracy results are obtained for voltage sag in ground faults. In [51] the fault position method has been used to determine an area of the power system in which faults result in voltage sag and affect the performance of the sensitive devices connected to the network. This concept has been used to introduce “predicted voltage sag abundance,” corresponding to the occurrence probability of these faults in the area of vulnerability. In [52,53], Monte-Carlo simulation employs normal probability distribution functions to determine fault resistance. In [54], voltage sags are considered to be stochastic. Therefore, the results are obtained as probability distributions of occurrence that can be used to extract mean values. Finally, the voltage sags are analyzed statistically to predict the outage of devices.

2.2.3. Arc and high impedance fault prediction

High energy arc faults in MV boards might interrupt processes and threaten personnel and devices. High energy arcs are mainly caused by bad connections and insulation degradation [55]. The energy of such faults depends on factors, including current, voltage amplitude, and arc duration [56]. To reduce the occurrence of arcs as a result of accidental contacts, training staff, insulated buses, and improved maintenance can be used to prevent such faults [57]. Studies at the end of 80 s showed that open-ended coaxial type connector is used as a sensor to measure permittivity or dielectric of biologic material [58,59]. These sensors are similar to D-dot sensors in terms of operation principle and structure. In [60], D-dot sensors have been presented to measure high voltages and fast-front, including the waste voltage of surge arresters. In [61], a differential electric field sensor (D-Dot) has been presented to monitor the electric board with air insulation online. Also, Discrete Wavelet Transform (DWT) has been used for noise depletion of partial discharge signals and transient states created as a result of an electric arc. DWT analyzes multi-resolution signals in time and frequency domains. The comparison between the primary signal and the noise depleted signals in time and frequency domains shows the superiority of DWT compared to infinite impulse response (IIR) in noise depletion of PDs and low power arcing signals.

In [62], a device for detecting the arc fault on a power line carrying a load current has been presented. This device monitors and analyzes the flowing load current through the line. The analyzed energy values have

been compared to several threshold values. To determine the occurrence of the arc fault on the power line, a number of energy values greater than the threshold have been used. In another patent, a system for arc fault detection in electrical power systems has been presented [63]. This system consists of a data collection unit and a computing device. This computing device has been programmed to: i) receive the first data from the data collection unit representing the transient behavior of the first signal, ii) eliminate the normal load component from the first data, and iii) determine whether an arc event is present on the electrical power system.

In [64], some problems have been evaluated such as evolving faults, DC decay, transients, inrush, clipping, and intermittent faults, and has analyzed various waveforms. Then, based on the phase currents and currents in symmetrical components, it has presented a criterion for detecting the fault type or system health. In another part of this report, the effects of the arc voltage on various waveforms have been evaluated, and the arc fault has been detected using the arc voltage estimation method.

A practical method has been presented in [65] for detecting high-impedance faults in distribution feeders. This system studies the behaviors of the electrical variable and high-impedance faults using several algorithms and input data in an on-line manner. Then, the probability of the occurrence of a fault in the system is computed using a smart processing system.

A patent in 1996, an on-line expert detection system has been presented for detecting high-impedance faults in a distribution network connected to an AC power source [66]. This system consists of an apparatus that detects high-impedance faults based on an expert's knowledge of the behavior of this type of fault and the efficiency of various fault detection techniques.

2.2.4. Overvoltage fault prediction

In [67], the prediction of temporary overvoltage in the radial MV distribution network generated after a single phase to ground fault has been studied. This reference shows that neutral voltage for MV distribution networks might exceed 1pu (a maximum of 2.5p.u.) at low resistance faults and crucial parts of overhead lines. Transient overvoltage on healthy phases might be larger than $\sqrt{3}$ p.u. (a maximum of 3.5p.u.). Fault currents are also affected and reach two times the values calculated by simple methods, i.e., ignoring the series impedances. In [67], analytical formulas have been presented for predicting the maximum overvoltage, fault currents, and critical fault distances. Also, they are validated using ATP-EMTP simulations.

2.2.5. Transient stability fault prediction

One of the effective methods of reducing global outages or cascading failures is the accurate and fast prediction of transient stability. Authors of [68] have presented a transient stability prediction method by combining trajectory fitting (TF) and extreme learning machine (ELM). An ELM-based method has been utilized in the central station to reduce the time interval of the prediction process. While the TF-based method has been used in the local station to ensure accuracy. Besides, data corruption has been considered to ensure that the proposed algorithm is robust. Fig. 7 compares this method's performance with prediction methods based on TF and ELM and the method based on SVM. The results indicate the superiority of the hybrid method [68].

2.2.6. Insulator fault prediction

Predicting fault in the network components is vital because it may cause a fault in the whole electric network. In [69], a new wavelet group method of data handling (GMDH) is presented to predict fault in power insulators. The insulators' surface may contain contamination so stuck that rain could not naturally remove the contamination and cause short circuits in severe weather conditions. Therefore, there could be a way to predict the probable time that a fault may happen. An ultrasound is a

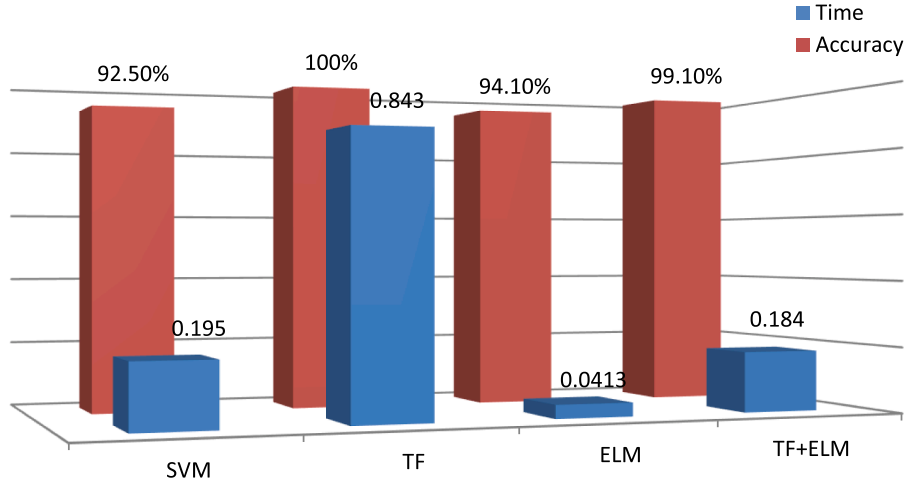


Fig. 7. Comparing the transient prediction methods in terms of accuracy and response time [68].

device that generates an audible time series-based noise that is applied to detect possible failures. The wavelet transform has been used to extract all sampled data that is generated by ultrasound devices. These features are fed to GMDH for time-series prediction and/or long-short term forecasting. For each specific set of inputs ($x_i, i = 1, \dots, n$), the predicted output ($\hat{y}$) is determined using (11):

$$\hat{y}(x_1, \dots, x_n) = a_0 + \sum_{i=1}^m a_i f_i \quad (11)$$

where f_i , a_i , and m are elementary functions, coefficients, and the

number of function components. The flowchart of fault identification and forecasting is shown in Fig. 8. This method allows us to prevent a fault in insulators before it happens and increase network reliability.

2.2.7. Transformers fault prediction

Transformers are one of the most important pieces of equipment in the distribution network which have the task of changing voltage level. By predicting faults in transformers, power outages in microgrids and distribution networks can be prevented. Also, financial losses due to power outages and complete failure of transformers can be minimized by timely detection of a fault in this equipment. In a report from EPRI, a

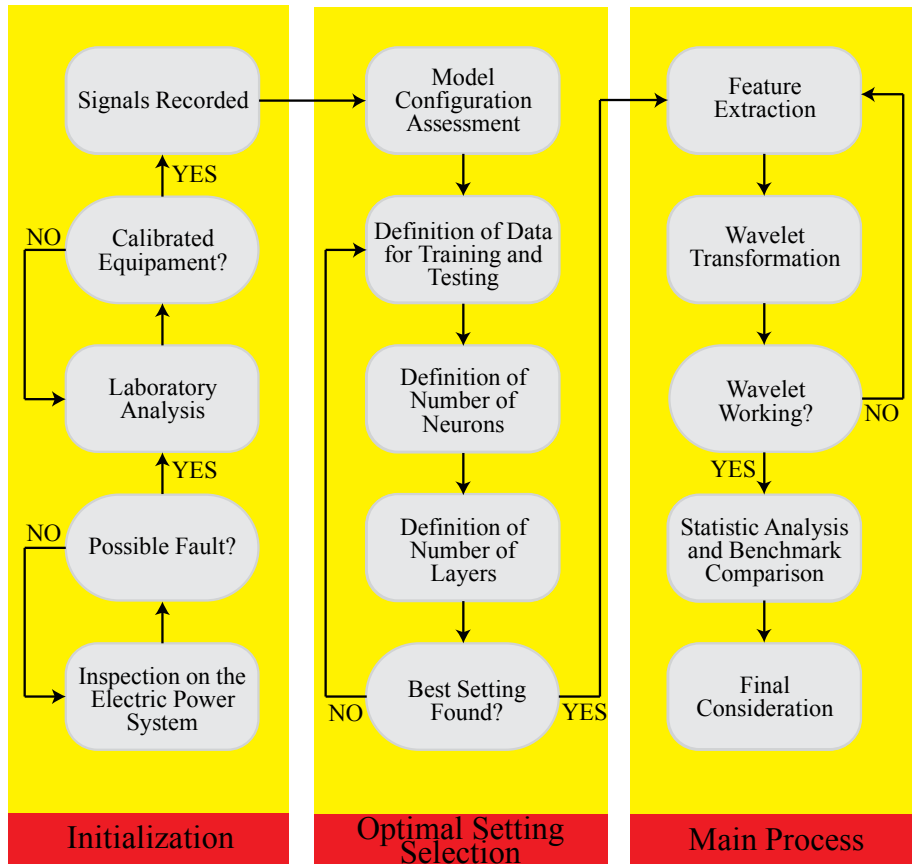


Fig. 8. Flowchart of insulator fault identification and forecast analysis [69].

large number of actual fault data, incipient faults, and operational problems with power system apparatus (such as wind-induced conductor slap, failing external transformer bushing, internal failure of transformer primary bushing, etc.) have been collected to predict and prevent the occurrence of faults. This project has been carried out by 14 companies on about 60 feeders from 14 substations [70]. For instance, one part of this project relates to the fault of the internal winding of the transformer. All the voltage and current data have been recorded one week before the occurrence of this fault. These data showed that 5 transient faults have occurred during one week, ultimately leading to a power outage. Fig. 9 shows one of the first measurements. In Fig. 9(a), the measured load current of the substation was about 105 RMS amperes, and failure precursors caused intermittent five- to ten-ampere increases. The waveforms corresponding to the same episode are shown in Fig. 9(b).

In summary, these data can be employed to predict faults as follows:

- Announcement of a developing problem: no other apparatus shows the problem and no complaint has been made by the customer.
- Locating the problem: locating the fault can be difficult even if a fault is intermittent, not causing a persistent outage, and even if the feeder is long and geographically dispersed.
- No complaint from customers: This incipient failure is detected and repaired without any event causing the customer to complain at any time.
- Prevention of persistent outage: These frequent faults cause momentary disruptions in an interval but do not cause a persistent outage. Frequent faults do not resolve on their own. They can be hidden for a considerable amount of time. However, they ultimately occur and increase damaging effects and cause a permanent outage.

Gaussian mixture models are widely used for estimation, clustering, and extracting features [71,72]. In [73], a method is presented to predict faults in transformers based on the hidden markov model of dissolved gases analysis. Gaussian mixture model is used to extract the feature from the fault data. Transformer operating modes are divided into three types: healthy, sub-healthy and faulty. The flowchart of the algorithm for updating the model parameters is shown in Fig. 10.

In this work, by observing and predicting the trend of the characteristic of gases $\text{CH}_4/\text{H}_2(=R_1)$, $\text{C}_2\text{H}_2/\text{C}_2\text{H}_4(=R_2)$ and $\text{C}_2\text{H}_4/\text{C}_2\text{H}_6(=R_3)$ fault can be classified and predicted. For instance, for any value of R_1 and $R_2 < 0.1$ and $R_3 < 0.1$, the operation of the transformer is normal. For the case of $R_1 > 0.1$, $R_2 < 0.1$ and $R_3 > 3$, the thermal problem has happened and the temperature of transformers' oil has been exceeded of 700 degrees of centigrade.

2.2.8. Line and busbar fault diagnosis

In [22], fault diagnosis in electrical power systems has been performed based on the Bayesian network's approach. The tree-model is

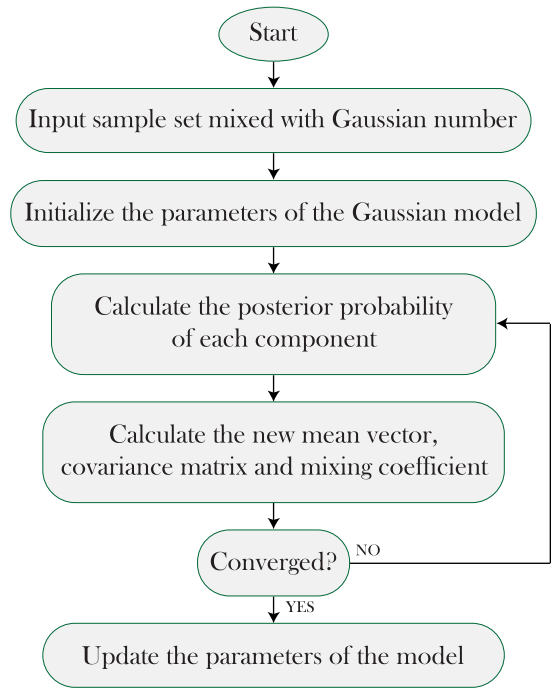


Fig. 10. The flowchart of Gaussian mixture model [73].

applied for detecting the fault in the busbar because of its flexibility in nature regardless of data uncertainty or incompleteness. A brief explanation of a simplified Bayesian network with Noisy-And and Noisy-Or nodes, which are the basis of the element-oriented models for faulty estimation sections in power system networks, is as follows:

- Noisy-Or model:

The belief degree of N_j for the sets of conditional N_i nodes that have associated inhibitory influence probability to N_j of q_{ij} each is obtained as follows:

$$Bel(N_j = True) = 1 - \prod_i (1 - c_{ij} Bel(N_i = True)) \quad (12)$$

where $c_{ij} = 1 - q_{ij}$ show the degree to which N_i causes N_j . A Noisy-Or node operates like a conventional logic connector OR with a simple difference of not implying necessarily true if only one of its conditions is true.

- Noisy-And model:

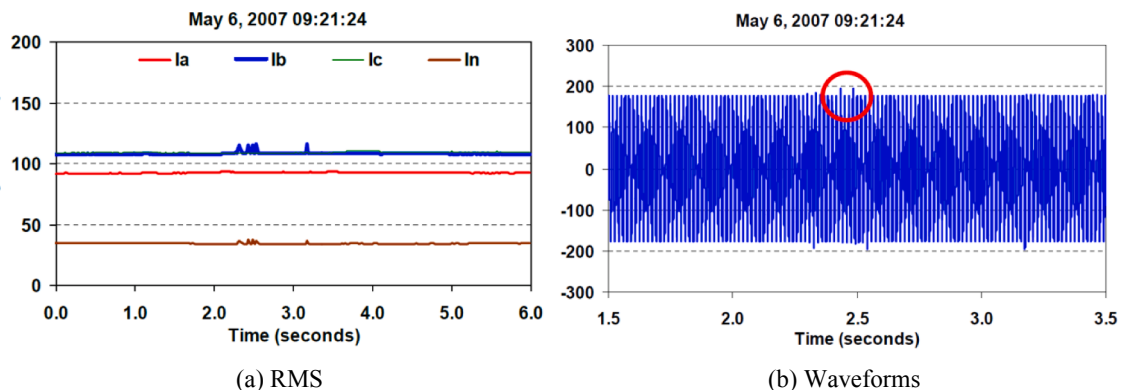


Fig. 9. Precursors one week before transformer failure [70]

The belief degree of N_j for the sets of conditional N_i nodes which have associated causing influence probability to N_j of q_{ij} each is obtained using (13):

$$Bel(N_j = True) = 1 - \prod_i (1 - c_{ij} Bel(N_i = True)) \quad (13)$$

where $c_{ij} = 1 - q_{ij}$ show the degree to which N_i refuse N_j . A Noisy-And node operates like a conventional logic connector AND with a simple difference of not implying necessarily false if only one of its conditions is false.

- The model for line fault:

The fault in line L_2 is depicted in Fig. 11. The circuit breaker and busbar are represented by CB_x and B_x , respectively. With the assumption of fault in L_2 , the relays located at the beginning and end of each section could operate individually. These relays in the networks can be divided into three kinds of main protection relay without time delay consist of pilot, first zone distance, and zero-sequence current protection, primary and secondary backup protection relay. The operation of these relays in the power system, their coordination with each other, and the circuit breakers hold the fault diagnosis model for fault in the section ($B_2 - B_3$) as depicted in Fig. 12.

The model's parameters are determined by a learning method like the error backpropagation algorithm, which is used to train a multilayer feedforward neural network. This method needs the power system's topology data, alarming and status data of all isolators, and circuit breakers from power plants and substations.

2.2.9. Short circuit and grounded fault classification and diagnosis

In [28], feature extraction, detection, and classification of fault in the DC network have been done by applying wavelet transform multi-resolution analysis methods with artificial neural networks. Parseval's theorem is used for pre-processing and feature extraction steps because of its robustness against collected signals' noise. In the following, a glimpse of wavelet-based multi-resolution analysis, feature extraction, and artificial neural network application for the case of fault detection and classification in DC networks is presented:

- Wavelet theory:

Wavelet transform is used for analyzing the high-frequency component (transient incident) of recorded power signals considering various scales of its coefficients. Continuous-time wavelet transform $W(a, b)$ with scale and transient factor a and b of a signal $x(t)$ for a given mother wavelet $\psi(t)$, is as follows:

$$W(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t) \psi\left(\frac{t-b}{a}\right) dt \quad (14)$$

Note that the formula, as mentioned above, usually is rewritten in its discrete form for engineering problems. The multi-resolution analysis is a framework to represent the different scales and hierarchical form of a signal. The wavelet transform-based multi-resolution analysis has the task of illustrating a signal $f(t)$ with respect to wavelet and scale functions. The following formulations are the representation of a signal $f(t)$ regarding the scaling $\phi(t)$ and wavelet $\psi(t)$ functions, and their

conjugates $(\bar{\phi}(t), \bar{\psi}(t))$.

$$f(t) = \sum_k c_{J,k} \phi_{J,k}(t) + \sum_{j=1}^J \sum_k d_{j,k} \psi_{j,k}(t) \quad (15)$$

$$c_{J,k} = \sum_k f(t) \bar{\phi}_{J,k}(t) \quad (16)$$

$$d_{j,k} = \sum_k f(t) \bar{\psi}_{j,k}(t) \quad (17)$$

As can be seen from Fig. 13, by passing a signal $x[n]$, which is the sampled version of continuous signal $f(t)$, through two high and low pass filters, wavelet and scaling coefficients, which represent the high and low-frequency component of the input signal, are computed to extracting the feature of recorded faulty current for detection and classification purpose. The criteria for selecting the mother wavelet (decomposition scales and wavelet function) should be chosen properly because these parameters impact the calculation burden and precision of the fault detection method.

- Feature extraction and classification:

The output of wavelet multi-resolution analysis (approximation and detailed coefficients) cannot be used directly as the classifier input because of the need for reducing feature dimension. Besides, the existing feature extraction methods, such as standard deviation, RMS, mean, and Shannon-entropy, which are among statistical approaches, are not robust against noise. Therefore, an energy-based method, namely Parseval's theorem, which is robust against noise, could be a key solution. The energy of the fault signal can be written in terms of the orthonormal basis of wavelet and scaling functions which is shown in (18)

$$E_{c_j} = \sum_{i=1}^{q_j} |c_{ji}|^2, \quad E_{d_j} = \sum_{i=1}^{q_j} |d_{ji}|^2 \quad (18)$$

The vector for feature extraction can be selected from the difference of energy held in the wavelet coefficients at each scale between normal signal and faulted signal. The mathematical representation of feature vector x is as follows:

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_j \\ \vdots \\ x_J \end{bmatrix} = \begin{bmatrix} E_{Fault,d_1} \\ \vdots \\ E_{Fault,d_j} \\ \vdots \\ E_{Fault,d_1} \end{bmatrix} - \begin{bmatrix} E_{Normal,d_1} \\ \vdots \\ E_{Normal,d_j} \\ \vdots \\ E_{Normal,d_1} \end{bmatrix} \quad (19)$$

Features of grounded and short circuit fault in DC and AC bus have been extracted by applying a wavelet transform-based multi-resolution analysis algorithm with db10 and Parseval's theory. The extracted feature vectors are the input of an artificial neural network (ANN) for training and testing the determined model's performance. The structure of the selected neural network is depicted in Fig. 14.

To demonstrate the operation of the system and to figure out if it is normal or faulty, two binary output is considered with the following description:

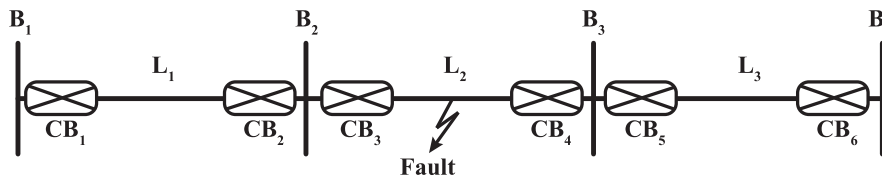


Fig. 11. The schematic diagram for a line fault [22].

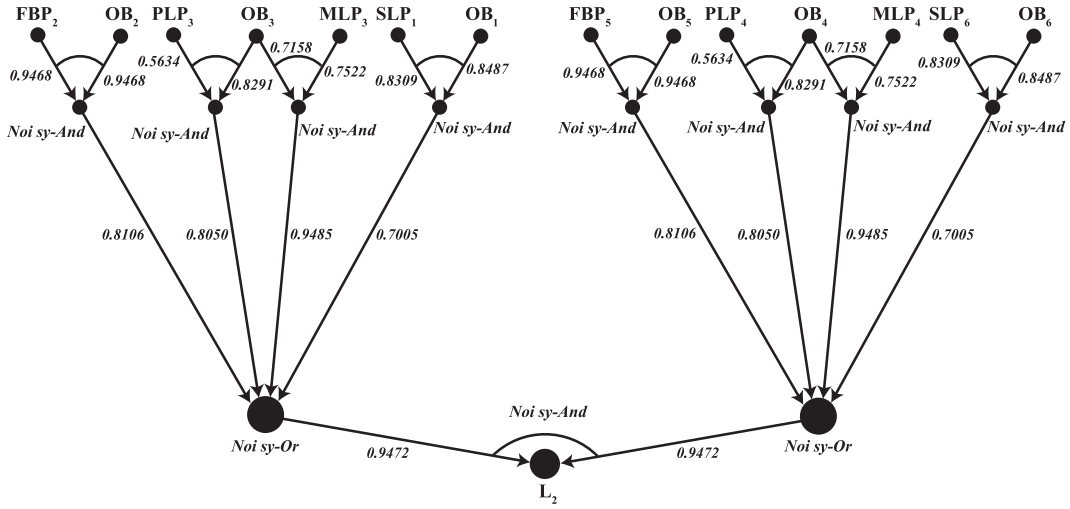


Fig. 12. A tree model for diagnosing faulty section of distribution line [22].

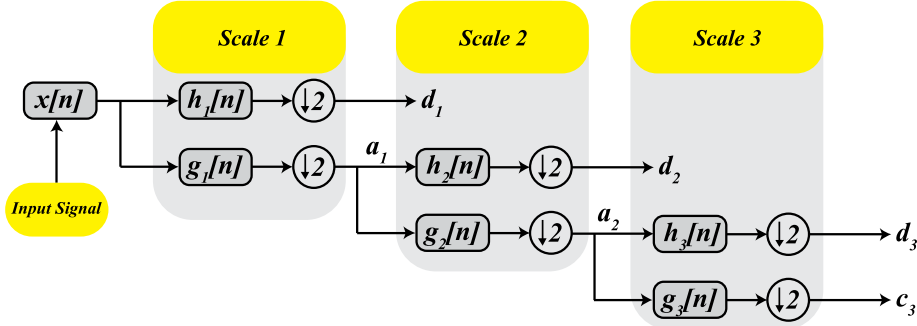


Fig. 13. The structure of wavelet transform with high and low pass filters [28].

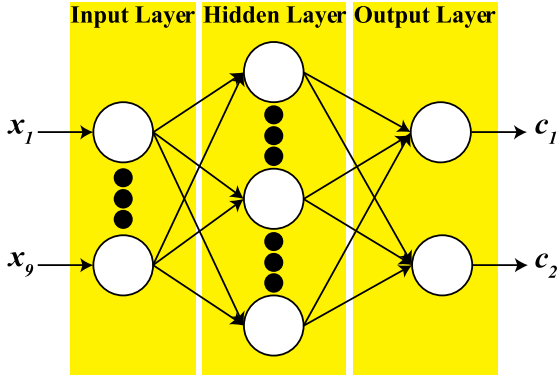


Fig. 14. Structure of a three-layer feedforward neural network [28].

$$\begin{cases} C_1 = 0, C_2 = 0 \rightarrow \text{Normal Operation} \\ C_1 = 1, C_2 = 0 \rightarrow \text{Short Circuit Fault on DC bus} \\ C_1 = 0, C_2 = 1 \rightarrow \text{Short Circuit Fault on AC side} \\ C_1 = 1, C_2 = 1 \rightarrow \text{Ground Fault in Zone Three} \end{cases} \quad (20)$$

Fault classification in distribution networks equipped with distributed generations has been done using a convolutional neural network (CNN) [74]. The raw sampled data of current and voltage are used as an input of CNN, which has several advantages of no need for pre-processing steps, converting input data to a grayscale image, and feature engineering step. This method can classify all types of fault and even no-fault conditions as an output of the model. The block diagram of fault classification using CNN is shown in Fig. 15.

2.3. A combined results and discussion for fault prediction methods in electrical distribution networks

In this part, many state-of-the-art methods for fault prediction in distribution networks have been reviewed. It can be seen that as the main tool for fault prediction, machine learning methods have been employed to predict the fault in the distribution network. The most widely used methods for predicting faults are expert systems, Bayesian neural networks, LSTM networks, and SVM. Since this field is still in its infancy and there are not many reported results and references, it is not possible to find a comprehensive method that can be used to predict the types of faults. In addition, because each method is specifically used to predict one type of fault, it is not possible to examine the disadvantages and advantages of these methods. However, the authors tried to show the accuracy and brief comparison between different methods using Fig. 4 and Fig. 7. Our investigation clearly shows that fault prediction although offers great benefits, needs further research to be employed widely in the electrical industry.

3. Fault location in electrical distribution networks

Electrical distribution networks have various sections and branches. Thus, fault location not only has to determine fault distance like transmission line but also must locate the faulty section. The distribution systems differ from the transmission systems regarding the following [3,4]:

- Presence of load branches on the main feeder and outside the main feeder.

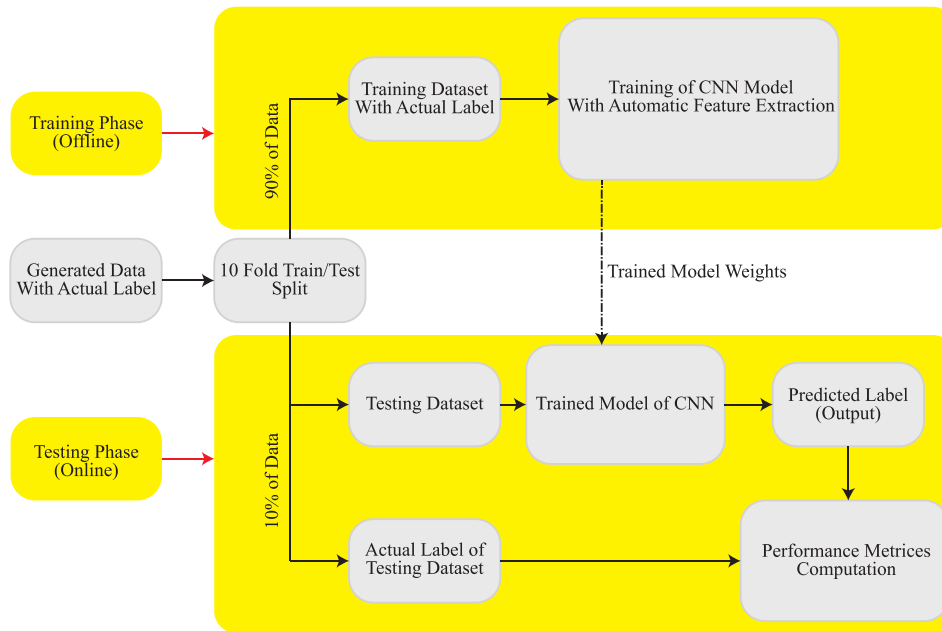


Fig. 15. Block diagram of the CNN-based fault classification [74].

- Heterogeneity of the feeder due to the presence of the underground cables and overhead lines with different characteristics.
- Imbalance because the lines are not displaced.
- Imbalance due to the presence of single-phase, two-phase and three-phase loads.
- Measuring current and voltage values at the beginning of the feeder.

In this part, determining fault distance, identifying the faulty section, and the relevant studies in electrical distribution networks are investigated. In addition, fault location methods in the presence of DG, smart grid and microgrid are reviewed separately.

3.1. Fault distance determination in electrical distribution networks

The faulty lines should be detected, repaired, and connected again in minimum time to improve the restore the electricity supply. A lot of methods have been presented to determine fault distance in the distribution networks. In the past, the power networks were protected by mechanical relays, and accessing voltage and current data was not possible. Therefore, fault location was done through operators and line-by-line patrolling. Furthermore, in this way, the use of expert knowledge in locating faults in feeders is very important and suitable. In this traditional practical method, the line was first divided into two or more parts by opening the installed power switches along the PDS line. Then, they do switching from the beginning of the feeder to detect the fault part. So, if it is interrupted it means that the fault is in the section between the substation and the first opened power switch else the fault is in other sections after the first power switch. This process should continue till the fault section is found by the maintenance group. This action was repeated in all feeders to determine the point and location of the fault. It should be noted that in places where the line did not have power switches in the middle of the line, the operators would open the jumpers and perform the same actions as mentioned before. Then, with the advancement of technology, power switches such as reclosers and sectionalizers were installed along with the feeders and using them and the relevant performance information, the faulty section is located faster than before. Today, fault locators such as the sel400 entered transmission lines. Furthermore, the installed distance relays have a fault locator that is only used on transmission lines. Moreover, today, some reclosers such as ABB, Tavrida, Entech, have a fault location part. This

fault location part is just accurate for straight lines without laterals and branches and needs special load conditions. Some fault indicators are installed at feeders. When a fault occurs at downstream of these indicators, they turn on the flasher and send a message for the operator to know that fault has happened. Furthermore, other methods that use an alarming ampere meter, automatic oscilloscope, have been presented; but these methods have not been widely used because they require primary operation and skills in addition to being time-consuming and inaccurate. Therefore, these practical methods cannot solve the fault location in power networks. The methods which are recently presented for fault location in power systems measure voltage and current on one side or both sides of the line to calculate the fault distance. These methods are classified into the following three classes [3,75]:

- Methods based on the main frequency of the voltage and current signals (impedance methods)
- Methods based on differential equations.
- Methods based on transient waves and high-frequency components of the voltage and current (Traveling waves).

3.1.1. Determining fault distance using impedance based method [3,75,76]

The first group includes the methods that employ main frequency components (50 or 60 Hz) of the voltages and currents, line parameters, and load information to estimate the fault location. Thus, the impedance seen from the beginning of the feeder is calculated, and then the fault distance is estimated using a heuristic algorithm. These methods are cheap and simple; but, they have to deal with the multiple response problem (in DNs).

• Determining the Equivalent Load at the End of each Section

In this part, considering Fig. 16, calculating the impedance of equivalent load at the end of each section is described.

If a fault occurs between bus i and bus j , the equivalent impedance at the end of each section (the equivalent impedance at bus j) should be determined as follows:

$$Z_{Leq} = (Z_1^{-1} + Z_2^{-1} + Z_3^{-1} + Z_{Lj}^{-1})^{-1} \quad (21)$$

Impedances Z_1 , Z_2 , and Z_3 are determined through the calculation of

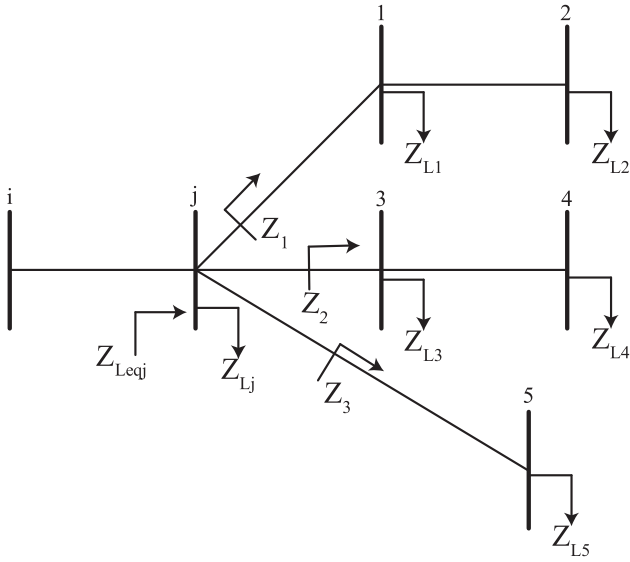


Fig. 16. Single-line diagram of the sample DN.

the equivalent impedance seen from the j^{th} node. The load impedances and line impedances are put either in series or in parallel to calculate these impedances. It should be mentioned that the distributed line model is used for each section to obtain higher accuracy, as shown in Fig. 17.

Eqs. (22), (23), and (24) show how the equivalent impedance of each section connected to node j is determined.

$$Y_1 = \left(\left[\left(\left(Z_{L_2}^{-1} + \frac{Y_{12}'}{2} \right)^{-1} + Z_{L_2}' \right)^{-1} + \frac{Y_{12}'}{2} + Z_{L_1}^{-1} + \frac{Y_{j1}'}{2} \right]^{-1} + Z_{j1}' \right)^{-1} + \frac{Y_{j1}'}{2} \quad (22)$$

$$Y_2 = \left(\left[\left(\left(Z_{L_4}^{-1} + \frac{Y_{34}'}{2} \right)^{-1} + Z_{L_4}' \right)^{-1} + \frac{Y_{34}'}{2} + Z_{L_3}^{-1} + \frac{Y_{j3}'}{2} \right]^{-1} + Z_{j3}' \right)^{-1} + \frac{Y_{j3}'}{2} \quad (23)$$

$$Y_3 = \left(\left[\left(\left(Z_{L_5}^{-1} + \frac{Y_{j5}'}{2} \right)^{-1} + Z_{L_5}' \right)^{-1} + \frac{Y_{j5}'}{2} \right] \right) \quad (24)$$

• Determining Voltage and Current at the Beginning of each Section

Using the information recorded at the beginning of the feeder (voltage and current) and considering that each node is at the downstream of the node at the beginning of the feeder, the voltage of the downstream node (voltage and current) and its input current from the previous node (V_{kj}) can be obtained using (25) and (26).

$$V_{kj} = k_0 + k_1 l_{ij} + k_2 l_{ij}^2 + k_3 l_{ij}^3 + k_4 l_{ij}^4 + k_5 l_{ij}^5 \quad (25)$$

$$I_{kij} = k_{f0} + k_{f1} l_{ij} + k_{f2} l_{ij}^2 + k_{f3} l_{ij}^3 + k_{f4} l_{ij}^4 + k_{f5} l_{ij}^5 \quad (26)$$

in which:

l_{ij} : is the length of section $i-j$.

$k_0 - k_{f5}$ coefficients: are described entirely in [77].

In [77], Eq. (27) is formulated to obtain the fault distance for single-phase, two-phase, and three-phase to ground faults. In each fault type and number of phases involved with fault, the number of terms existing in $\sum$ is determined. When a fault occurs, k_{dm} coefficients are determined using current and voltage at the beginning of each section, and fault distance is obtained using the improved impedance method calculated using (27).

$$x^5 \left[\sum_{m \in p} \text{Im}(k_{5m} I_{Fm}^*) \right] + x^4 \left[\sum_{m \in p} \text{Im}(k_{4m} I_{Fm}^*) \right] + x^3 \left[\sum_{m \in p} \text{Im}(k_{3m} I_{Fm}^*) \right] + x^2 \left[\sum_{m \in p} \text{Im}(k_{2m} I_{Fm}^*) \right] + x \left[\sum_{m \in p} \text{Im}(k_{1m} I_{Fm}^*) \right] + \left[\sum_{m \in p} \text{Im}(k_{0m} I_{Fm}^*) \right] = 0 \quad (27)$$

Also, (28) has been presented for the two-phase fault between phases A and B [77]. The other two-phase faults have similar equations.

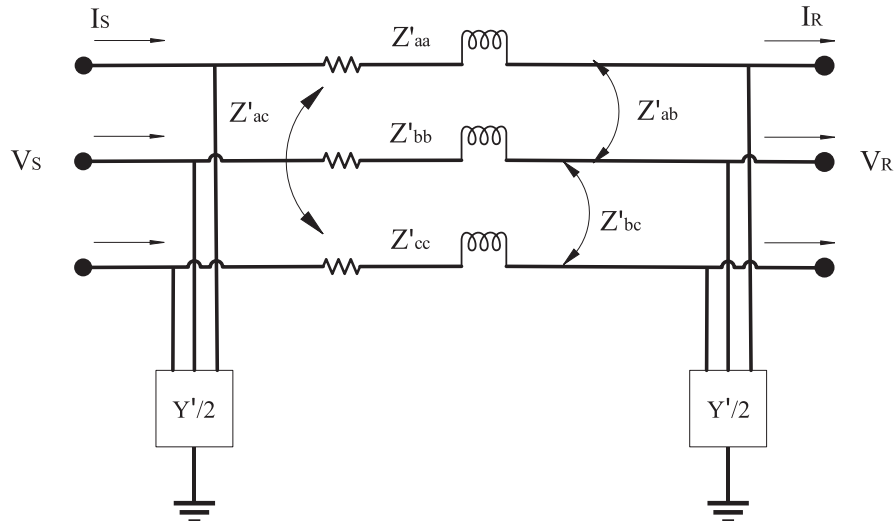


Fig. 17. Circuit model of each section of the DN.

$$\begin{aligned}
& \text{Im}\{(k_{5a} - k_{5b}) \cdot I_{F_a}^*\} x^5 + \text{Im}\{(k_{4a} - k_{4b}) \cdot I_{F_a}^*\} x^4 + \text{Im}\{(k_{3a} \\
& - k_{3b}) \cdot I_{F_a}^*\} x^3 + \text{Im}\{(k_{2a} - k_{2b}) \cdot I_{F_a}^*\} x^2 + \text{Im}\{(k_{1a} - k_{1b}) \cdot I_{F_a}^*\} x + \text{Im}\{(k_{0a} \\
& - k_{0b}) \cdot I_{F_a}^*\} \\
& = 0
\end{aligned} \quad (28)$$

In [78], ten different impedance-based methods have been investigated in terms of using the information before and after fault time, load model, line model, presence or absence of branches in the DN, considering system imbalance, considering system heterogeneity, and additional information. Also, various methods have been simulated, and their advantages and disadvantages have been presented. The results presented in this reference show that the higher fault resistance and the longer the fault distance, cause an increase in the estimation error.

In [79], an iterative algorithm has been presented to find fault distance in radial distribution systems. In this reference, the currents and voltages calculated at the beginning of each branch and the dynamic load model are used to achieve higher accuracy. The presented method in [79] locates all fault types. In this reference, the effects of different fault resistances, fault inception angles, load variation, and fault distances on the accuracy of the proposed model have been examined. The simulation results in [79] have been obtained using data from an actual distribution system in Southern Brazil. Currently, this formulation is used as software in the CEEE-D distribution operations center. The maximum error of this method in three-phase to ground fault and 100 Ω fault resistance, considering the distributed parameter line model, is 10.97%. In this reference the main formula for determining faults' location and impedance is as follows:

$$\begin{bmatrix} x \\ R_F \end{bmatrix} = \frac{1}{M_{1m} I_{F_m i} - M_{2m} I_{F_m r}} \begin{bmatrix} I_{F_m i} & -I_{F_m r} \\ -M_{2m} & M_{1m} \end{bmatrix} \times \begin{bmatrix} V_{S_m r} \\ V_{S_m i} \end{bmatrix} \quad (29)$$

where i and r indices represent the imaginary and real part and $m \in \{a, b, c\}$. V_{S_m} , x and I_{F_m} are upstream voltage, fault distance, and fault current, of a faulty section respectively. M_{1m} and M_{2m} are two calculated parameters which are determined using the following sums:

$$M_{1m} = \sum (Z_{mk_r} I_{S_{k_r}} - Z_{mk_i} I_{S_{k_i}}) \quad (30)$$

$$M_{2m} = \sum (Z_{mk_r} I_{S_{k_i}} - Z_{mk_i} I_{S_{k_r}}) \quad (31)$$

A single line diagram of fault is depicted in Fig. 18. The fault location algorithm is as follows:

1. Recording pre-fault load current and using as post-fault load current
2. According to Fig. 18 fault current can be calculated as:

$$I_{F_m} = I_{S_m} + I_{R_m} = I_{S_m} - I_{L_m} \quad (32)$$

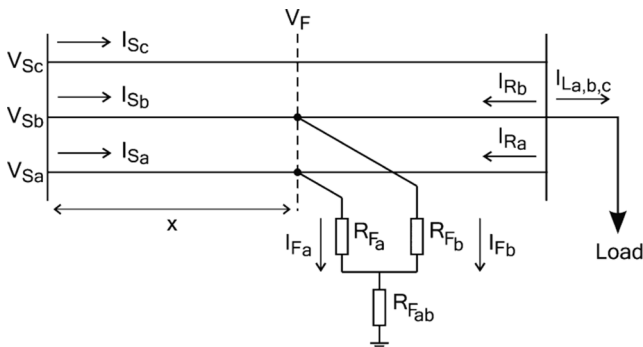


Fig. 18. Single diagram of fault [79].

3. The location of fault can be determined using Eq. (29).

4. Fault current is updated using the following equations:

$$\begin{bmatrix} V_{F_a} \\ V_{F_b} \\ V_{F_c} \end{bmatrix} = \begin{bmatrix} V_{S_a} \\ V_{S_b} \\ V_{S_c} \end{bmatrix} - x \cdot \begin{bmatrix} Z_{aa} & Z_{ab} & Z_{ac} \\ Z_{ba} & Z_{bb} & Z_{bc} \\ Z_{ca} & Z_{cb} & Z_{cc} \end{bmatrix} \begin{bmatrix} I_{S_a} \\ I_{S_b} \\ I_{S_c} \end{bmatrix} \quad (33)$$

$$Y_{pq} = [(I - x)Z_{pq} + Z_{L_{pq}}]^{-1} \quad (34)$$

$$I_{L_m} = [Y_{ma} \ Y_{mb} \ Y_{mc}] [V_{F_a} \ V_{F_b} \ V_{F_c}]^T \quad (35)$$

Firstly the voltage of fault point, then admittance of load from fault point of view and finally load current are calculated using Eqs. (33)–(35).

Check the following convergence condition

$$|x(\alpha) - x(\alpha - 1)| < \sigma \quad (36)$$

If the condition is satisfied, stop.

Authors of [80] have presented an algorithm considering the presence of cable lines in distribution networks, which has extended the impedance methods considering the effect of parallel capacitances in the model. This reference is used the current and voltage information at the beginning of the feeder and the dynamic load model to determine fault distance. With this difference that the parallel admittance of the studied section is assumed to be equal to the total parallel admittance of the section. Then, the capacitive current and the load current obtained using the pre-fault condition, and the initial current of the studied section is used to determine the fault current. Next, using the calculated distance for fault location, new reactance, and current of the capacitive branch and the new load current are calculated, and the algorithm is repeated. This algorithm is continued until the difference in the calculated distance of the current and the previous states becomes lower than a pre-determined value. The flow-chart of this algorithm is depicted in Fig. 19.

Various reports have been presented to study and locate faults by EPRI. For example, in [81], the Takagi, arc-voltage, current phasor only, current magnitude only loop reactance, and current-profile methods were used in one feeder from a distributed network to estimate the location of actual faults that occurred due to capacitor cutout and lightning arrester. These faults are shown in Fig. 20 and Fig. 21. It was eventually observed that the accuracy of these methods had decreased with an increase in the distance of the fault from the monitoring station.

Two fault location techniques have been tested in 3 actual distribution feeders in [82]. Reactance to Fault (RTF) is an impedance-based fault location technique that uses the substation voltage and current measurement. On the other hand, Voltage Drop Fault Location (VDFL) is a fault location technique based on voltage drop and operates based on distributed-voltage measurements. Long Island Power Authority (LIPA) and Hydro-Québec (HQ) hosted this project in coordination with EPRI. Evaluation of the fault location technique on the feeders of LIPA was only performed for the VDFL method, and its error was reported to be from –600ft to –700ft. The accuracies of both methods were reported to be very good in the HQ feeder. VDFL has been presented with an error percent of 1.76% or 2100ft, and RTF has been presented with an error percent of 2.59% or 3084ft. The results show that VDFL performs slightly better than RTF. This study shows that both methods are suitable for short urban feeders and long rural feeders.

In [83], an impedance-based method has been presented to determine fault distance in DN. In this reference, the model is considered for each section, and the presented modified impedance method improves accuracy. To enhance the preciseness of the fault location π line model is used for each section. After some simple mathematical operations two quadratic equations are acquired for both phase to phase and grounded faults, respectively as follows:

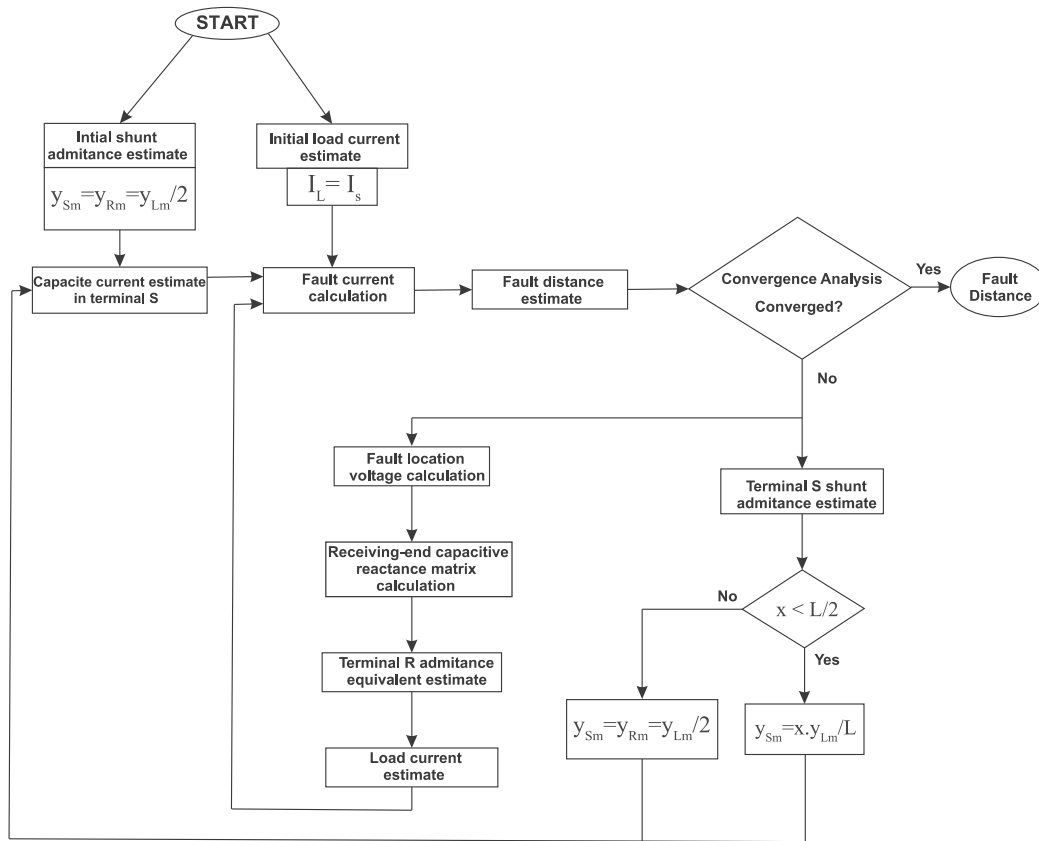


Fig. 19. The flow-chart of iterative fault location algorithm [80].

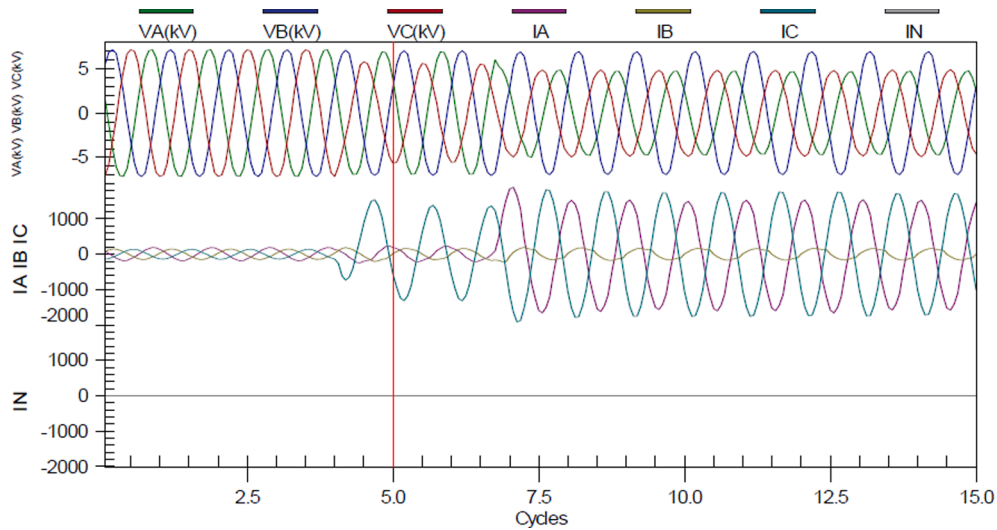


Fig. 20. Single line-to-ground fault in phase C evolving into phase A [81].

$$x^2 \cdot \left[0.5 \cdot \sum \text{Im} \{ M_k \times I_{F_k}^* \} \right] - x \cdot \left[\sum \text{Im} \{ N_k \times I_{F_k}^* \} \right] + \left[0.5 \cdot \sum \text{Im} \{ V_{S_k} \times I_{F_k}^* \} \right] = 0 \quad (37)$$

$$x^2 \cdot 0.5 \times \text{Im} \{ (M_i - M_k) I_{F_i}^* \} - x \times \text{Im} \{ (N_i - N_k) I_{F_i}^* \} + \text{Im} \{ (V_{S_i} - V_{S_k}) I_{F_i}^* \} = 0 \quad (38)$$

The matrices M and N contain line parameters. The maximum error for an IEEE 34-bus network is 1.58%, which means 1551 m, for the total network length (98,180 m).

Another method has been presented in [84], which determines the fault distance using the distributed parameter line model, current, and voltage at the beginning of the feeder. This method has presented different equations for determining fault distance, which are functions of current and voltage at the beginning of the feeder. This method is sensitive to load and fault resistance. The maximum error of this method in the fault resistance of 10 Ω is 1.11%.

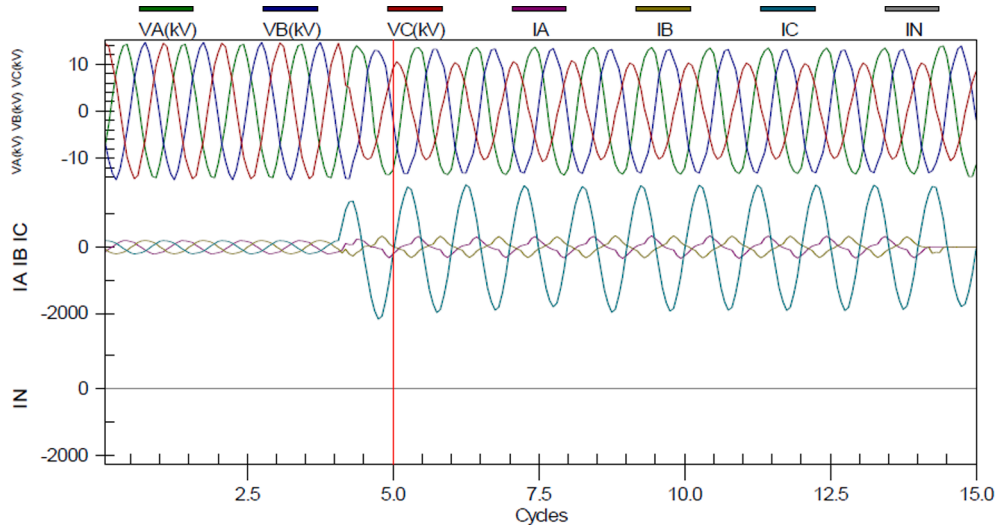


Fig. 21. Single line-to-ground fault in phase C [81].

In [85], fault location is determined in four-wire networks through improving the presented algorithm in [3]. All of these algorithms are iterative.

A non-iterative two-terminal fault location algorithm that employs only positive-sequence data is presented in [86]. The time-domain approach is used for its formulation. To extend the approach to untransposed lines, a modal transformation technique is used. A quartic equation considering synchronism mismatch has been expressed, and the angle of signal alignment mismatch is provided by its solution.

3.1.2. Fault distance determination based on differential equations

The second group includes fault location methods based on differential equations. These methods use line transition models. In some of these methods, fault resistance can be considered as an additional parameter and estimated concerning the fault point voltage. Although considering and ignoring fault resistance changes the equations a little, it does not change the basics of the methods; thus, resistance is the fault is considered to be zero. The differential method has some disadvantages, among which requiring communication links with high bandwidth, requiring a high sampling rate, fast and accurate data synchronization can be mentioned. These methods are classified into two classes, including the differential equation method based on the lumped line model and the distributed parameter line model. The differential equation method based on the lumped line model of the line calculates the line resistance and inductance between the locator and fault point using the measured time samples. The above method is sensitive to the integration of the sampling rate and signal spectrum. Although increasing sampling rate increases estimation accuracy, numerical problems are created. Also, this method is not appropriate for long lines [87,88]. In the distributed parameter line model, the line parameters are distributed along the line uniformly. This model considers the parallel capacitances of the lines also. Therefore, the accuracy of the method based on the distributed model is highly dependent on the sampling rate constraints. Consequently, accuracy is sensitive to the selected data window [89].

- *Fault distance determination based on differential equations methods using lumped line model (AC and DC)* [87].

Considering the lumped model of the line, if x_1 and x_2 are two points on the line, (39) can be used to represent the relationship between phase currents and voltages in a three-phase line.

$$V_{x_1}(t) - V_{x_2}(t) = (x_2 - x_1) \left[ri(t) + l \frac{di(t)}{dt} \right] \quad (39)$$

The above equation can be rewritten as follows using the sequence conversion:

$$V = x_f Ri_A + x_f L \frac{di_B}{dt} \quad (40)$$

Voltages and currents of phases are comprised depending on the fault type. Using the digitalized samples of voltage and current, Eq. (40) is extended concerning $x_f R$ and $x_f L$ using a proper differential approximation. Thus, by obtaining a set of equations, the unknowns $x_f R$ and $x_f L$ are obtained. According to Eq. (41):

$$x_f Ri_A + x_f L \frac{di_B}{dt} - V = \varepsilon \quad (41)$$

$x_f R$ and $x_f L$ should be determined such that the error function ε is minimized. These equations can be solved using different methods, including metaheuristic algorithms, mathematical methods [87].

Knowing the voltage and current information at both terminals and considering Fig. 22, fault location can be calculated for short DC lines based on Eq. (42) [90].

$$x = \frac{u_{rec}(DC) - u_{inv}(DC) - R i_{inv}(DC)}{R(i_{rec}(DC) - i_{inv}(DC))} \quad (42)$$

where:

$u_{rec}(DC)$ and $u_{inv}(DC)$ are the DC components of voltage at terminals of the rectifier and the inverter. Also, $i_{rec}(DC)$ and $i_{inv}(DC)$ are the current components at the terminals of the rectifier and the inverter.

- *Fault Distance Determination Based on Differential Equations using the Distributed Parameter Line Model (AC and DC)*

In the distributed parameter line model, the following equations are used:

$$-\frac{\partial V_P(x,t)}{\partial x} = R I_P(x,t) + L \frac{\partial I_P(x,t)}{\partial x} \quad (43)$$

$$-\frac{\partial I_P(x,t)}{\partial x} = G V_P(x,t) + C \frac{\partial V_P(x,t)}{\partial x} \quad (44)$$

where: V_p and I_p are voltage, and current of phases, and R , L , G , and C are 3×3 matrices associated with series resistance, inductance,

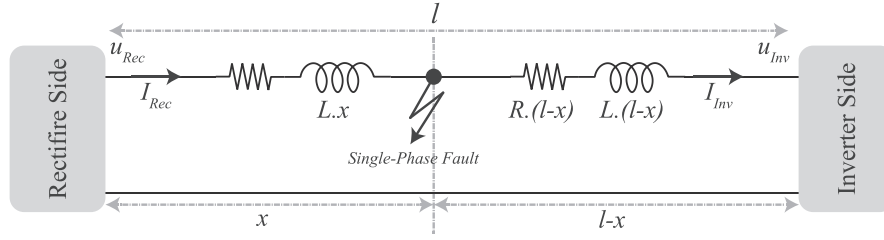


Fig. 22. Short line model when a fault occurs.

conductance, and capacitance where G is usually ignored in the overhead lines [87]. Using modal transformation, fuzzy variables can be calculated for systems with un-transposed lines, and the high-frequency transients can be used for calculating the fault distance in the minimum time.

Considering Fig. 23 and knowing the voltage and current information at both terminals, fault location can be calculated for DC distributed line based on Eq. (45) [90].

$$x = \frac{u_{rec}(DC) - u_{inv}(DC) - Rl i_2(DC)}{R'(i_1(DC) - i_2(DC))} \quad (45)$$

In the above equation, x is the distance of the fault from the rectifier terminal, $u_{rec}(DC)$, and $u_{inv}(DC)$ are DC components of the voltage at terminals of the rectifier and inverter. $i_{rec}(DC)$ and $i_1(DC)$ are the currents at the terminal of the rectifier, and $i_{inv}(DC)$ and $i_2(DC)$ are the currents at the inverter terminal. R' is the resistance per unit length of the line.

3.1.3. Traveling waves methods

The third group includes methods operating based on traveling waves. Traveling waves methods were first used in the 1950s for Fault location. However, due to high cost, low reliability, and maintenance problems, they were put aside in the 1970s. With the development of technology, digital signal processing techniques, and the possibility of using accurate GPS systems, fault location methods based on traveling waves emerged again. Traveling waves are the current and voltage waves generated at the beginning of fault at the fault location and propagated towards the line terminals. These waves move in overhead lines with light speed and damp gradually at the terminals and fault locations due to the damping effect and wave reflection laws. The methods within this group are fast. However, they are complicated, and their implementation is difficult. The accuracy of these methods depends on line parameters, the accuracy of the measurement devices, and bandwidth [91].

Traveling waves-based fault location methods are performed based on required measurements in one or two terminals of the power networks. These methods are divided into 5 classes based on being active or passive. In the passive methods, the transients resulting from fault are used for location. The active methods inject a signal to the lines after resolving the fault by a key, and the generated transients are used to determine the fault distance [92,93].

Type A is an algorithm in which only one side of the line is sampled. In this method, the time required for the traveling wave resulting from

the fault event to go from the terminal to the fault location and return is measured. This method is passive because no signal is injected into the line.

Consider Fig. 24, assuming that measurements are performed at terminal M and the time difference between T_{M1} and T_{M2} is considered as Δt , by knowing the wave speed in the line (9) which is equal to the light speed, the fault distance from terminal M can be calculated as follows (T_{M1} and T_{M2} are the times at which the first and second traveling waves arrive from line to terminal M):

$$x_M = \frac{1}{2} \vartheta \Delta t \quad (46)$$

Type B is a two-terminal algorithm. In this method, the time difference of arrival of the wave resulting from fault occurrence to two terminals of the line is calculated. The required relationships for calculating fault distance can be obtained as follows:

$$x_M = \frac{1}{2} [\vartheta(T_{M1} - T_{N1}) + L] \quad (47)$$

$$x_N = \frac{1}{2} [\vartheta(T_{N1} - T_{M1}) + L] \quad (48)$$

In the above equations, T_{N1} is the time at which the first traveling wave resulting from fault arrives at terminal N , and L is the line length,

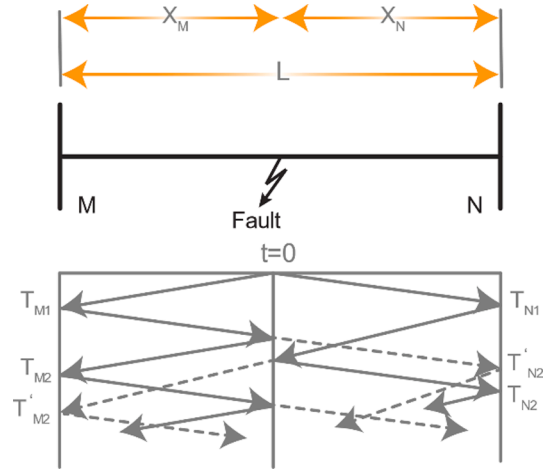


Fig. 24. Traveling wave propagation diagram.

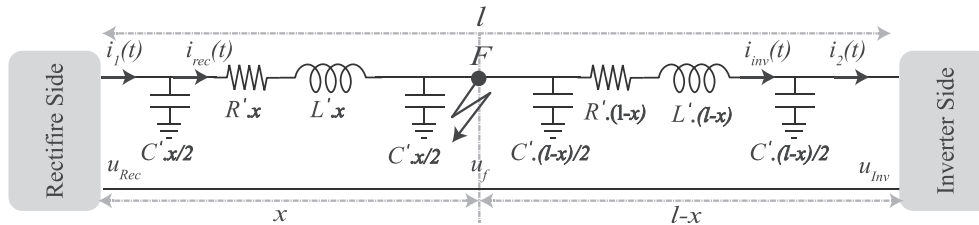


Fig. 23. Fault occurrence on the two-terminal line in the distributed line model.

X_M , and X_N are the distances of the fault from terminals M and N . This method is passive also because the transients caused by the fault are used for calculations.

Type C employs measurements of a terminal, and it is similar to type A. This method is active. Because it uses the transients caused by signal injection to the line after line outage by the switch to determine the fault distance instead of using the transients caused by the fault. Type D is similar to type B and to operate correctly, and requires measurements of both terminals to be done simultaneously. This method is active. The relationships given for type B can be used to measure fault distance in this method. Type E employs measurements of one terminal and the transients generated when the switch electrifies the line to calculate the fault distance.

In [94], a method is presented based on correlation functions and traveling waves to determine the faulty section. This method presents a criterion and specifies a standard range for the presented criterion. When a fault occurs, this criterion is calculated separately for each section. If the calculated criterion of a section is smaller than the defined value, the section is introduced as the faulty section.

A fault location approach based on a traveling wave for two-terminal lines with unsynchronized current measurements from (IEDs)⁴ captured at both ends is presented in [95]. The only information needed to use as an input to the algorithm is the velocity of the line propagation and the arrival time of the 1st and 2nd traveling wave. The IED hardware, the error of data synchronization, and the delay of software processing are taken into account for this algorithm formulation. The first two fault location points are calculated employing the peak arrival times of two traveling waves seen at two terminals of the line in the presented method. Next, the faulted half-section information is used to choose the correct location of the fault. Analyzing the first traveling wave at two terminals of the line, the faulted half section has been specified. This approach is sensitive to load switching and transformer switching.

3.1.4. Advantages and disadvantages of fault distance determination methods in electrical distribution networks

Considering the studies reviewed in the fault distance determination part, the pros and cons of these methods are summarized in Fig. 25.

3.2. Fault section estimation and hybrid methods in electrical distribution networks

Various methods have been presented for fault section estimation in distributed networks. On the one hand, artificial intelligence methods can estimate both the fault distance and the fault section estimation. On the other hand, some methods (fault indicators and Magnetic sensors, protection coordination and current analysis, and state estimators methods) can only estimate the fault section. All of these methods can complement the methods of determining the fault distance to design a hybrid method (a method is used for fault distance determination and a method is used for fault section estimation). These methods include:

- Methods based on magnetic sensors and fault indicators.
- Methods based on protection coordination and current analysis.
- Methods based on state estimators
- Methods based on Artificial intelligence.

In the following, the studies on fault section estimation in DNs are reviewed.

3.2.1. Magnetic sensors and fault indicators

In recent years, researchers have been studying magnetic sensors for monitoring power systems. These sensors include those with anisotropic magnetoresistance [96], those with giant magnetoresistance [97] along

with other sensors [98,99]. These sensors are capable of measuring the electric current straight from the magnetic field in the environment. These advanced sensors have been proven to be applicable to transient analysis in power system applications involving both medium and high voltages [100,101]. They can gather reliable data for detecting, classifying, and locating faults in power systems since they are free from CT saturation along with other types of measurement errors that affect the accuracy of a number of techniques. These sensors are applied similarly to distribution and transmission networks. Hence, the techniques employed in transmission lines are equally valid for distribution lines. [102] has introduced a fault location method that leverages non-contact measurements of magnetic fields and the autoregressive model of the magnetic signature taken from the surroundings of the power system. This strategy is unable to self-adjust to handle usual changes in the magnetic fields. The researchers in [103] consider the magnetic field strength signal to be a THIF⁵ signature. In this method, the fault location criterion consists of the particular trend between the phases of high-frequency magnetic field components during a single period. This criterion assumes that the spatial changes in the magnetic field's high-frequency components appear in the magnetic field vectors' phase shifts at the sensing point. The algorithm presented to determine the exact distance between the THIF fault and the sensing point is shown below:

1. Determining the data window of the magnetic field strength's x and y high-frequency components
2. Calculating the high-frequency components of the generated magnetic field strength in the XY-plane
3. Determining the phase shift of the magnetic field strength's high-frequency components for every sample individually in the obtained data window
4. Calculating the total phase shift
5. Determining the distance between the fault and the sensing point according to the total phase shift

Each higher and lower envelopes of high frequency information of the x component are determined. The mean of all pre-determined high and low envelopes are calculated as follows:

$$m_{i,j}(t) = \frac{(u_{i,j}(t) + l_{i,j}(t))}{2} \quad (49)$$

The first frequency component determines using the following iterative equations:

$$H_x(t) - m_{1,1}(t) = h_{1,1}(t) \quad (50)$$

$$h_{1,k-1}(t) - m_{1,k}(t) = h_{1,k}(t) = H_x(t) - (m_{1,1}(t) + m_{1,2}(t) + \dots + m_{1,k}(t)) \quad (51)$$

$$c_1(t) = h_{1,k}(t) = H_x(t) - (m_{1,1}(t) + m_{1,2}(t) + \dots + m_{1,k}(t)) \quad (52)$$

where c_1 is the first frequency component. The second frequency component can be determined in the same way. The stopping condition of iteration is reached when only one extremum remains. The faulty feeder identifies using the energy of both components of the signal using the quantile regression method [104].

According to Fig. 26 fault types can be determined, if the measured magnetic field satisfies Eqs. (53)–(58) for phase A to G, phase B to G, phase C to G, phase A to phase B, phase A to phase C, and phase B to phase C faults, respectively:

$$\frac{B_y}{B_x} \approx \frac{\sin\theta}{\cos\theta} = \tan\theta \quad (53)$$

⁴ Intelligent Electronic Devices

⁵ Tree-Related High Impedance Faults

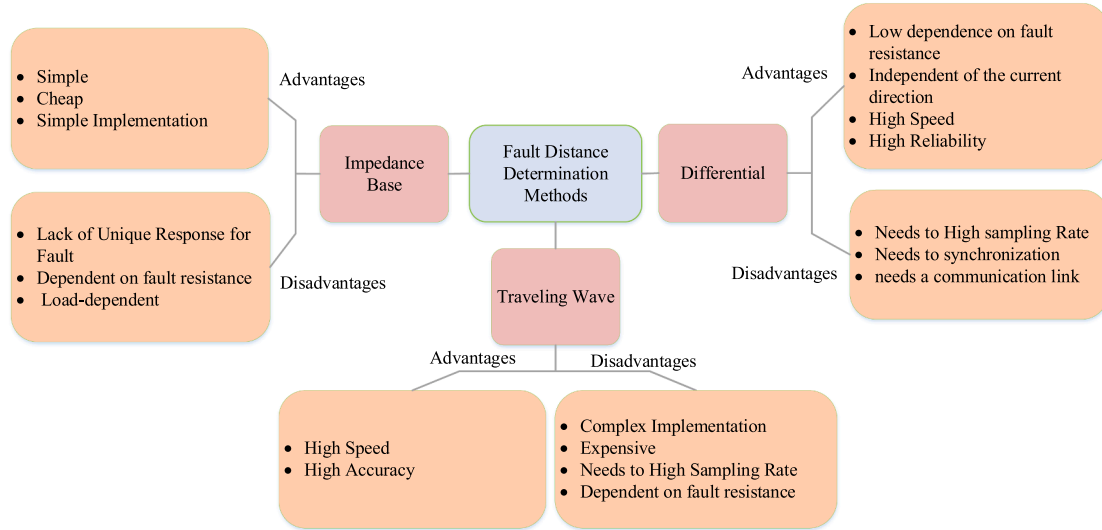


Fig. 25. Pros and cons of the fault distance determination methods.

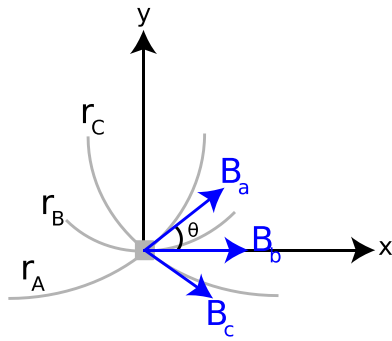


Fig. 26. Magnetic field component at the measurement point [104].

$$B_y \approx 0 \quad (54)$$

$$\frac{B_y}{B_x} \approx \frac{-\sin\theta}{\cos\theta} = -\tan\theta \quad (55)$$

$$\frac{B_y}{B_x} \approx \frac{\frac{1}{r_A} \sin\theta}{\frac{1}{r_B} + \frac{1}{r_A} \cos\theta} \quad (56)$$

$$B_x \approx 0 \quad (57)$$

$$\frac{B_y}{B_x} \approx \frac{-\frac{1}{r_C} \sin\theta}{\frac{1}{r_B} + \frac{1}{r_C} \cos\theta} \quad (58)$$

Finally, the exact location of the fault can be determined using the following equation:

$$d = 0.00062\phi_T + 0.36 \quad (59)$$

The aforementioned equation is estimated using curve-fitting on the total phase shifts at various locations.

The authors in [105] have presented a magnetic-field-based non-contact fault localization technique and have used it to locate SC faults in 11 kV distribution lines. This technique requires the deployment of magnetic field sensor modules at the substation and on selected distribution poles. This sensor module is capable of effectively localizing SC faults according to levels of the magnetic field measured on various branched main feeders. The mathematical model of magnetic field theory is derived from the Biot-Savart law and describes its effects in balanced and unbalanced load circumstances. The magnetic field

waveform of triangular, vertical and horizontal configurations produced at single-phase to ground fault. The mathematical presentation of the magnetic field component is given as follows:

$$\vec{B} = [B_b + (B_a \cos\theta_1 + B_c \cos\theta_2)i_x + (B_a \sin\theta_1 - B_c \sin\theta_2)i_y + 0i_z] \quad (60)$$

$$\vec{B} = \left[\frac{\mu_0 I_b}{2\pi r_b} + \left(\frac{\mu_0 I_a}{2\pi r_a} \cos\theta_1 + \frac{\mu_0 I_c}{2\pi r_c} \cos\theta_2 \right) i_x + \left(\frac{\mu_0 I_a}{2\pi r_a} \sin\theta_1 - \frac{\mu_0 I_c}{2\pi r_c} \sin\theta_2 \right) i_y + 0i_z \right] \quad (61)$$

In the aforementioned equation, B_a , B_b and B_c represent magnetic field components which are generated by phase-a, -b and -c. The magnetic field sensor is linear and measures the magnetic field along its direction. In the case of a single three-phase circuit, its direction is horizontal and it could represent the triangular and vertical directions as depicted in Fig. 27.

The equations of three configurations are as follows:

$$B_{x, \text{horizontal}} = \left[\left(\frac{\mu_0 I_b}{2\pi y_{i2}} \right) + \left(\frac{\mu_0 I_a}{2\pi} \times \frac{x_{j1}}{(x_{j1}^2 + y_{i1}^2)} \right) + \left(\frac{\mu_0 I_c}{2\pi} \times \frac{x_{j2}}{(x_{j2}^2 + y_{i3}^2)} \right) \right] i_x \quad (62)$$

$$B_{x, \text{vertical}} = \left[\left(\frac{\mu_0 I_b}{2\pi} \times \frac{x_{j1}}{(x_{j1}^2 + y_{i2}^2)} \right) + \left(\frac{\mu_0 I_a}{2\pi} \times \frac{x_{j1}}{(x_{j1}^2 + y_{i3}^2)} \right) + \left(\frac{\mu_0 I_c}{2\pi} \times \frac{x_{j1}}{(x_{j1}^2 + y_{i1}^2)} \right) \right] i_x \quad (63)$$

$$B_{x, \text{triangular}} = \left[\left(\frac{\mu_0 I_b}{2\pi} \times \frac{x_{j1}}{(x_{j1}^2 + y_{i2}^2)} \right) + \left(\frac{\mu_0 I_a}{2\pi} \times \frac{x_{j2}}{(x_{j2}^2 + y_{i3}^2)} \right) + \left(\frac{\mu_0 I_c}{2\pi} \times \frac{x_{j2}}{(x_{j2}^2 + y_{i1}^2)} \right) \right] i_x \quad (64)$$

The flow-chart of fault localization is depicted in Fig. 28.

In [106], for radial distribution networks, a fault location algorithm is presented, employing the observers of synchronized and distributed voltage traveling waves. The approach presented is based on the time-stamping and the capture of the arrival time of a traveling wave

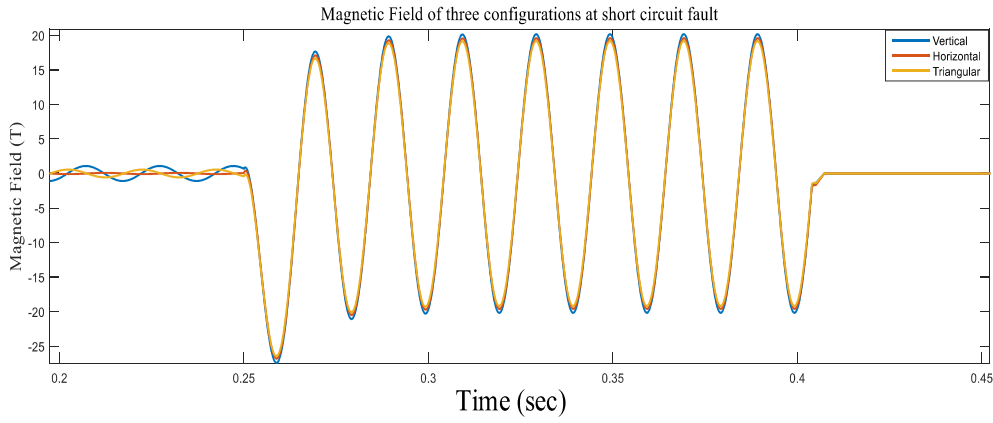


Fig. 27. Magnetic field waveform of vertical, horizontal and triangular configurations at single-phase to ground fault [105].

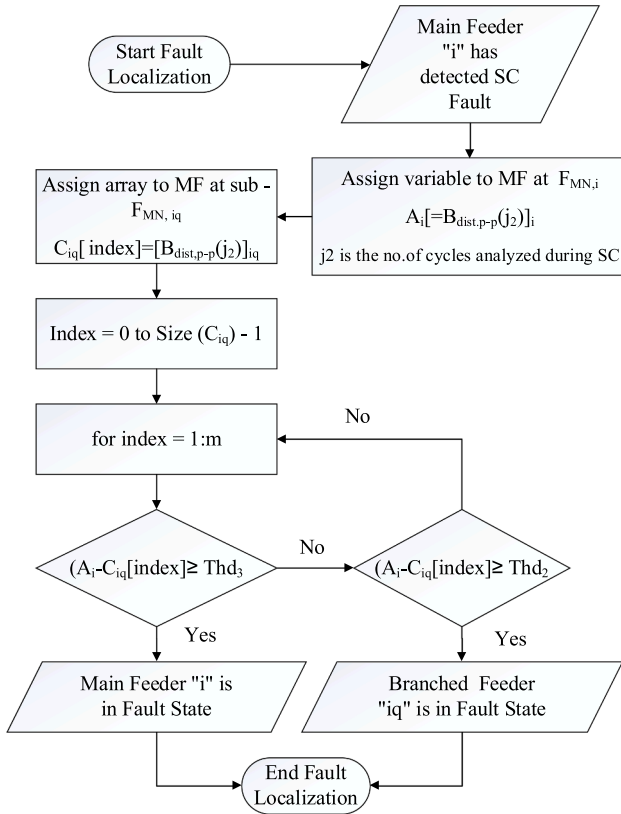


Fig. 28. The flow-chart of fault localization based on magnetic field [105].

(fault-induced) at a high frequency. The Monte Carlo method, an uncertainty analysis, is used to explain the robustness as well as the accuracy of the algorithm. For solving the faulted section localization issue in MV power distribution networks with asymmetrical fault, an approach is presented in [107]. It employs only voltage measurements such as voltage angle, magnitude, and sequence components obtained by sensing devices mounted at the low voltage side of step-down transformers, which are not needed to be time-synchronized. Ref. [108] has presented a non-contact fault location technique. This technique depends on magnetic fields generated from current signals that are measured via magnetoresistive sensors placed only on power line terminals under the first tower's phase conductors at the substation portals or at both terminals. It utilizes the Extended Kalman filter for processing these measurements. Moreover, this technique uses a traveling wave strategy for localizing faults.

In fault section estimation using fault indicators, a message is transmitted to the dispatching center by a sudden increase in current and a drop to zero of voltage, and the fault is reported [109,110]. In [111], a fault location system has been presented for an IEEE 34 bus network. This system has at least one sensor and one fault location evaluation system. This sensor is placed in the distribution network to measure the current and divide the distribution network into at least two areas. This system includes one fault area identification unit, which selects one of the areas in which the fault has occurred by calculating the fault current from the sensor.

A large number of sensor data is required by the distribution network. Consequently, using these measurements is uneconomical because of the distribution network's size.

3.2.2. Protection coordination and current analysis methods

In protection coordination methods, the fault section is determined based on the placement of various protection equipment in suitable locations and the introduction of a special protection coordination method capable of fault detection in every section using the current pattern at the beginning of the feeder [112,113]. In [76], the current and voltage recorded at the beginning of the feeder and impedance method are employed for determining the possible fault locations and determining the main fault section using the protection devices. The disadvantage of this method is that unique characteristics are not generated in each section's current pattern. Because the number of protection devices is limited, and there are a large number of distribution feeder branches. Also, the accuracy of this method depends on the fault resistance. The error of this method is 2.95% for a fault resistance of 50 Ω and 1% for faults with resistances smaller than 30 Ω . In current analysis methods, the faulty section is calculated using the fault current pattern and introducing a criterion [75,114]. The main problem in the methods is the interference in the detection of the main section of the fault. This problem occurs when the fault resistance is high and affects the current domain. finally, the performance of the relay is affected.

3.2.3. State estimators

In the state estimator-based methods, sufficient data is provided to estimate the faulty section using recorded data from a set of measurements, process monitoring, feedback control, parameter updating, and transient data matching [115]. The main challenge of fault location is that the number of measurements is limited, and the fault location algorithm implementation is difficult due to high computation volume, high implementation cost, and low reliability in the presence of DG with high penetration and a high unbalanced current and changes of the network topology. A state estimator should be used to overcome these constraints. The state estimator is executed online and estimated incorrectly and accessible measurements of the network to evaluate the system [116]. The state estimation process can be applied to a three-

phase power network with unbalanced loads and lines [117]. Also, this process uses the power or current injected into the network as pseudo-measurements or online measurements (previous or predicted information of the network) and the line data. Therefore, this method does not depend on the dynamic behavior of DGs and the Thevenin equivalent of the external network [118]. In addition, phasor measurement units (PMUs) have been used in DNs recently which could help the estimation [117,118].

In [119], the fault location is performed considering a small number of measurements and the distributed parameter line model. Fault location using positive current and voltage sequence equations before the fault is converted to an optimization equation. This method is robust against various factors, including cyber-attack and measurement errors. The presented algorithm automatically employs the least absolute value estimator to detect and eliminate faults and determine their location. Authors of [117] have assumed a PMU on all buses of the network, and active fault location is performed. This method is presented for online fault detection and location through computing parallel synchrophasor-based state estimators. Parallel estimators use PMUs data and obtain network states considering a hypothetical fault location at different points. In [120], a method based on state estimation is presented for fault location in DNs using measurements obtained from AMI. The errors in real or virtual measurements during normal operation can be suppressed using state estimation methods to present the best estimate of the system. Besides, they can detect large measurement errors known as bad data. This concept is extended under fault conditions. The fault is assumed to be an unknown and temporary load, considered as bad data. This study has used bad data detection techniques in a variable weight matrix to specify the fault location. In [115], an algorithm is presented for active DNs to detect short-circuit faults using the state estimator. This modified algorithm is the conventional version of the state estimator, and it is compatible with a fault condition. This presented algorithm locates fault after detecting it. To locate a fault, estimated states before fault and the voltage and current recorder after fault are used. In [121], a method based on a state estimator is presented for fault location. This method uses the voltage measured by a limited number of PMUs and the fault current in the presence of DG sources. In this method, permutation matrices and admittance matrices are constituted using pseudo-measurements. The error vector of each bus is obtained, assuming a fault in each bus and updating system states. Then, the Euclidean norm of the calculated error vector is calculated by the estimator for each bus as its error; the maximum error is determined by the faulty bus. In [122], a set of nonlinear current and voltage equations between measurements and the fault points is used to locate faults. An algorithm is presented to determine the minimum number of devices required to locate faults. At first, the algorithm reduces the search space to increase the location speed. Then, the fault location problem is converted to a linear least-squares problem regarding fault location to obtain the fault point.

3.2.4. Artificial intelligence methods

Artificial intelligence methods are used for determining the fault section and location in distribution networks. The conventional fault location methods use mathematical theories (including the differential equation, Fourier Transform) to find the fault location. In intelligent methods, tools such as neural networks, genetic algorithm, fuzzy logic, and machine learning algorithms are used. In these methods, fundamental frequency components of current and voltage are used as the input data. The requirements such as big data bank and updating information are its disadvantages [123–125].

In [126], the wavelet transform has been used to detect the fault and its type. Then, the neural network is trained based on the angle and magnitude of the fundamental frequency of voltage and current. Furthermore, it is trained by the magnitude ratio and angle difference of the fundamental frequency waveform and the third harmonics. By comparing the information, the real fault characteristics in the main fault section are determined.

In [127], current and voltage information at the beginning of the feeder and some other feeder points is used during a fault event to determine voltage and voltage sag at each node. In this method, it is assumed that fault might occur in each section and the fault current is found. It then calculates the voltage sag at nodes with measurement devices and compares the fault current with the measured one. A fault is detected, if the values agree, fault has occurred, and its distance is specified using the current and voltage at the beginning of the feeder.

In [128], SVM and K-nearest neighbor algorithms are used to reduce the multiple-response problem in fault distance determination for detecting the faulty sections. This method operates based on the line voltage difference, line current difference, and power changes before and after a fault at the beginning of the feeder.

A technique for localizing phase to ground faults in a distribution system is presented in [129]. Based on that, the energy spectrum is divided into various levels, and the transient voltage is decomposed by a wavelet filter. The signal that is decomposed in each level includes a specific percentage of energy which is dependent on the wavelet filter's bandwidth and the frequencies of the path characteristic. Afterward, a method based on a neural network is presented for a line fault locator of the distribution system. The candidate feature for training data in the artificial neural network is the percentage of energy in each level. Multiresolution analysis can help to find applicable features of a signal.

The multiresolution analysis involves symbolizing a function at different scales. For a given function $F(x)$, the basic notation of wavelet decomposition can be written as follows:

$$F(x) = \sum_m \sum_k d_{m,k} \psi_{m,k}(x) \quad m, k \in \mathbb{Z}. \quad (65)$$

In the aforementioned formulation, $\psi_{m,k}(x)$ is the mother wavelet. Eq. (65) can be approximated from the beginning start point as follows:

$$F(x) = F_0(x) + \sum_{m=0}^{\infty} \sum_{k=-\infty}^{+\infty} d_{m,k} \psi_{m,k}(x) \quad m, k \in \mathbb{Z} \quad (66)$$

The increase in the mother wavelet time dilation can bring about the decreasing of the frequency bandwidth of higher levels which allows wavelet transform to be as a multiresolution filter bank. The details of each resolution output can be calculated with the help of the following formula:

$$D_m(x) = \sum_k d_{m,k} \psi_{m,k}(x) \quad m, k \in \mathbb{Z}. \quad (67)$$

As a result, the energy details of each level can be computed as follows:

$$E_{Dm} = \int_{-\infty}^{+\infty} D_m^2 = \sum_k d_{m,k}^2 \quad m, k \in \mathbb{Z}. \quad (68)$$

Eq. (68) enables us to extract rich enough features for fault location purposes.

To begin the fault location process, a suitable training dataset is prepared. Then, an application for the appropriate ANN structure is selected. A multilayer feedforward network is presented to estimate the fault location. A multilayer perceptron of four layers (an input layer, two hidden layers, and an output layer) is considered. The hyperbolic tangent sigmoid transfer function and a linear transfer function are used for the hidden and the output layers respectively. In this reference, ANN uses 498 samples as inputs. To train the ANN, the back propagation technique with Levenberg-Marguardt method is used. After training, using an independent data set of fault scenarios are tested.

In [130–132], the magnitude and phase of voltage sag are calculated and stored in the data bank for the fault simulated at each node. Then, the voltage sag magnitude and phase are obtained according to the fault voltage information and compared with the data bank to extract the possible fault locations. Finally, a criterion is defined to specify the main fault location among the possible fault locations. In [130], the presented

method has been tested on a 25 kV feeder in an urban distribution network in Canada, and the maximum error of this method for fault resistances of 50 Ω has been reported to be 1.4%. Moreover, the effect of the DG on the fault distance accuracy has been presented, and it is observed that the error depends on the power injected from the DG and that the DG can reduce the accuracy of the method.

In [133], a method is presented to detect fault distance and section considering the high-frequency transient waves between the fault location and the beginning of the feeder. This work uses mode 1 of the frequency spectrum for the faults between phases and applies mode 0 of the frequency spectrum for the phase to ground faults. In this method, a criterion is used that introduces the difference between the main and the side frequency components for detecting the faulty section. After determining the faulty section through traveling waves, the fault distance is measured. The disadvantage of this method is that it requires an accurate data bank, and the side dominant frequency component might be detected incorrectly.

The outcome of a fault location and classification method study employing EMTP software is presented in [134]. The simulated data analyzed with an advanced signal processing method according to wavelet analysis to derivate beneficial information from the signals. Discrete wavelet transforms which is used in this work is written as follows:

$$DWT(f, m, n) = \frac{1}{\sqrt{a_0^m}} \sum_k f(k) h^* \left(\frac{n - ka_0^m}{a_0^m} \right) \quad (69)$$

After passing a signal through a wavelet transform different details appear according to the output filter's frequency scale. The frequency band of scale is defined as follows:

$$\text{Frequency band of scale } 2^m = F/2^{m+2} \rightarrow F/2^{m+1} \quad (70)$$

One of the main factors that determine the applicability of wavelet-based fault location methods in the real word distribution network is the choice of wavelet mother. Therefore, Daubichies' wavelet with 4db wavelet with 8 decomposing levels has been chosen. For feature extraction purposes, statistical relations such as mean, median, mode, skewness, correlation coefficients, and central moment are considered for obtaining a proper decreasing or increasing model as input of the neural network. Due to the choice of statistical relations, the standard deviation equation which is written in the following can be helpful.

$$STD(\nu) = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (\nu_i - \bar{\nu})^2} \quad (71)$$

After the fault type is recognized, the faulty phase and its type of non-grounded and grounded can be identified using standard deviation voltage and current phases. To locate the faults, the fuzzy logic system (FLS) and the artificial neural network (ANN) are employed in a real underground distribution system. The flowchart for the presented method is shown in Fig. 29.

For fault location and fault section identification in the distribution network, a hybrid 2-stage approach is presented in [135], which is based on the level-5 detail coefficients gained by Discrete Wavelet Transform decomposition employing db4 mother wavelet. Two indices can be calculated by the Discrete Wavelet Transform detail coefficients: Entropy Per Unit (EPU) and Wavelet Energy Spectrum Entropy (WEE). The training data in the artificial neural network model employ these indices for the fault location and fault section identification tasks, respectively. The artificial neural network model Comparison, which is trained by the WEE and EPU indices, is provided for parameters like the computation time, processor consumption, prediction accuracy, and memory consumption.

Raw signals cannot be used for feature extraction goals. Therefore, discrete wavelet transforms (69) with $a_0 = 2$ and WEE of the signal can provide energy information of a signal. The wavelet energy of a signal is

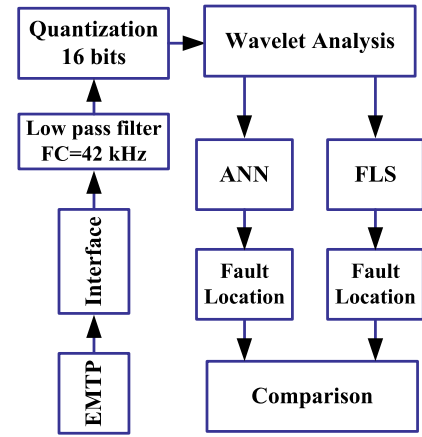


Fig. 29. The flowchart for fault location method using wavelet analysis, ANN, and FLS [134].

an absolute square of decomposition levels of that signal. Signal wavelet energy spectrum at j th scale and its wavelet energy can obtain as follows:

$$E_j = \sum_{k=1}^N E_{jk}, \quad j = 1, \dots, L \quad (72)$$

$$P_{jk} = \frac{E_{jk}}{E_j}, \quad j = 1, \dots, L \quad (73)$$

where: L is the number of decomposition level.

WEE and its per unit index are calculated as follows and use for the feature extraction and selection process.

$$W_{EEjp} = - \sum_k P_{jkp} \log P_{jkp}, j = 1, \dots, L \quad (74)$$

$$EPU_{jp} = \frac{- \sum_k P_{jkp} \log P_{jkp}}{(- \sum_k P_{jka} \log P_{jka}) + (- \sum_k P_{jkb} \log P_{jkb}) + (- \sum_k P_{jkc} \log P_{jkc})} \quad (75)$$

These extracted features are fed to the ANN model for training purposes to locate and identify the distance and faulty section. The designed ANN models are trained for all types of fault separately. To estimate the fault distance, these models take in sets of four inputs of EPU/WEE of the zero and three phase sequence line currents, and one output node. The sigmoid activation function and the linear activation function are employed in the hidden and output layer respectively. The following performance criteria are used to select the optimal ANN model: (i) Correlation coefficient (ii) Neural network size (iii) Testing with untrained datasets (iv) Error histogram. The untrained dataset accuracy is resolved by the percentage error of the ANN accuracy. The architecture of the ANN model designed for the fault location method is shown in Fig. 30.

A hybrid method is presented for locating short circuit single phase to ground faults in Ref. [4]. First, the improved impedance methods are used to specify the possible fault locations. Then, the faulty section is determined online using a voltage sag matching algorithm. This method has been tested on an actual distribution feeder in Iran. The error of this method has been reported to be 0.42% in the simulations.

3.2.5. Advantages and disadvantages of fault section estimation methods in electrical distribution networks

Considering the studies reviewed in the fault section estimation part, the pros and cons of these methods are summarized in Fig. 31.

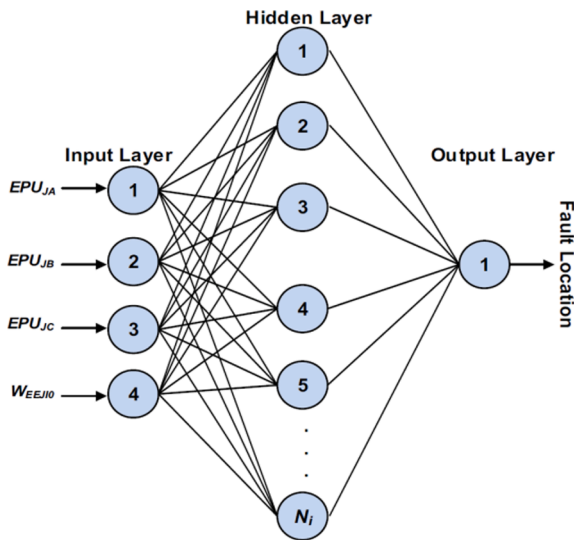


Fig. 30. Structure of the ANN model for the fault location method [135].

3.3. Fault location methods in DNs in the presence of DG resources

Due to the increase in electrical energy consumption in the recent decade, the transmission systems' current capacity has become irresponsive. Thus, to supply the demand and use renewable energies, various DGs are used in the DNs. A DG is a generating resource in the range of a few kilowatts to a few megawatts which can be connected and disconnected to the network at any time. DGs have different work scales based on their types (photovoltaic, wind turbines, fuel cell, etc.). Although DGs have many advantages, they cause protection problems in the DNs. Mainly, DGs inject a short-circuit current to the DN when a fault occurs that makes using fault location algorithms in radial DNs, difficult. Moreover, photovoltaic-based DGs output power is highly

dependent on the irradiation of the sun. Besides, the utility may not have adequate control over the placement and number of DGs which lead to high penetration in the distribution networks. Therefore, DGs behavior is not steady and can lead a network to islanded mode. Fault location in such a network is a challenging task. Impedance-based fault location algorithms are inapplicable when faced with high penetration DGs. In [136,137], the fault location method fails when a fault occurs between a substation and a low-capacity DG (kilowatt-DG) because of the low DG-side fault current. The gathered data of a network with DG is not rich enough to train a model properly (machine learning methods) when using intelligent methods because of their stochastic natures against fault. To overcome this difficulty a well-chosen pre-processing step is needed. Hence, fault location in the networks with DGs has its own difficulties that should be considered.

In [138], a high-frequency impedance-based fault location method is proposed, which is proper for systems with DGs. A short rectangle window with 6 ms fault transient states is used to prevent control loops with a cascade response time of about 10 ms. Also, the effect of fault resistance and fault starting angles have been considered.

In [139], a fault location method is presented using the self-repair concept in DNs in the presence of DGs. In this study, the fault location algorithm requires the transient and steady-state of the signals, the power flow algorithm, and the synchronization angle.

In [136], fault distance in DN in the presence of DGs is measured using an impedance method. In the presented method, current and voltage at the beginning of the feeder and DGs are measured, and the fault location is calculated by the π line model. The accuracy of this method is 98.5%. This reference includes an experimental test in the Clinical-Laboratory Center of Power System & Protection at Persian Gulf University. The error obtained in this test was reported to be 0.9273%, indicating the high accuracy of this method.

In [140], a fault location approach is presented, which is based on the energy analysis of the zero-sequence current (transient) in the selected frequency band (SFB), by the use of a methodology based on coordinated measurements in DNs. An equivalent network is used to

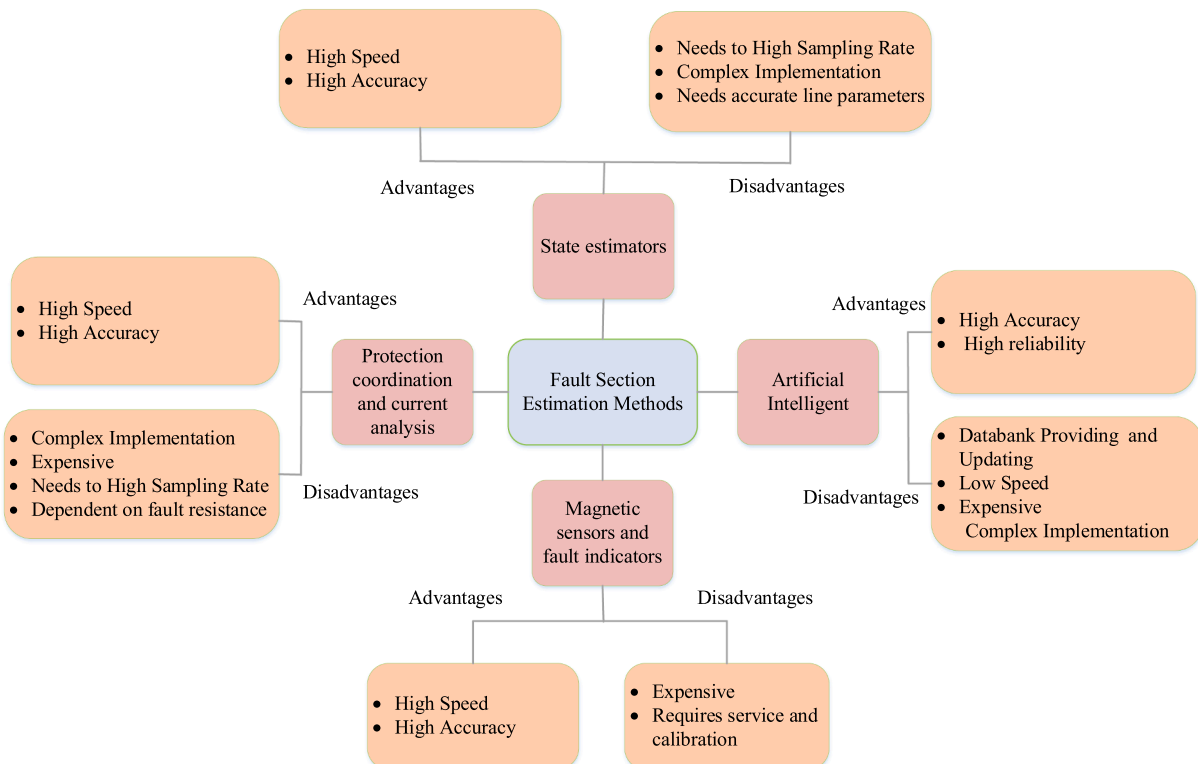


Fig. 31. Pros and cons of the fault section estimation methods.

study the equivalent impedance of a distribution network containing lateral branches. Moreover, the phase-frequency properties of the equivalent impedance are evaluated. In the SFB, the faulty line section transient energy that is greater compared with the energy (transient) of the healthy ones is established. A combined fault section estimation criterion is presented, and the execution scheme is explained using distribution level phasor measurement units.

In [141], a fault location method has been proposed for a single-phase to ground fault through synchronous measurement in active DN's in the presence of DGs. In this method, the fault location method includes 2 steps. First, the faulty section is estimated based on the fault characteristic model and information of PMUs. Then, the current and voltage of the phases of the estimated sections and a criterion for fault location are calculated. In this step, the Fibonacci algorithm is used to search for the exact fault location. The maximum error of this method for resistances of 100 Ω is 0.767%, and its response time is 0.847 s.

3.4. Comparison of different references in fault section estimation and fault distance methods in electrical distribution networks

Table 2 shows various fault location methods in DN's based on the DN's inherent characteristics and their impact on the accuracy of these methods. In addition, a comparison of properties, advantages, and disadvantages in different methods of fault location in DN's are shown in Table 3.

3.5. Fault location methods in smart grid and microgrid

o Fault Locating Challenges in Smart grids and Microgrids Compared to Conventional DN's.

The previous sections studied the fault location methods in conventional DN's. Considering the development of the DN's, distance determination and fault location methods have changed. The most challenging problem in the smart grid and microgrids protection section is the integration of renewable energy resources resulting in two-sided

power flow and various fault current levels in grid-connected or islanded modes. The fault current component is several times the rated current regarding the variable source type. Line impedance in DC microgrids is very small; therefore, fault current deviation is very large, and the fault current rises to hundreds of amperes in a few milliseconds. The problems mentioned above result in incorrect fault classification. Also, if the microgrid is islanded, the fault current component resulting from the energy resources might not reach the peak-up level of the protection device for fault detection; therefore, the sampling rate of the sensors should be high, the communication system should be very fast and reliable. The fault location methods should detect faults fast, reliably, and accurately considering the implementation of sensors, communication, and control systems. All of the above are the challenges of fault location methods. These challenges are given in the following:

- Because of DGs in the smart grids and microgrids, the power flow and fault current are from each DG's side which makes the fault location procedure a challenging task
- Fault location algorithms must well perform in both islanded and grid-connected modes
- Fault current levels in the microgrid is different for islanded and grid-connected modes which complicate fault location procedure
- Different control algorithms are applied to inverter-based DGs that has an effect on fault current and voltage
- Energy storage systems can be connected and disconnected to the network randomly in every possible node which brings about a robust fault location algorithm
- Many uncertainties in line parameters, loads, and a huge amount of data for processing can cause many difficulties
- o Communication and Automation Standards in Smart grid and Microgrid

Smart grids use communication sensors to collect information and send it to the control center. This information is used for fault detection, fault location, fault prediction and several other applications including automation, energy management, AMI, etc. Advanced communication technologies are needed to send this information. Various standards are

Table 2
Comparing various fault location methods and their characteristic in DN's.

Reference	[83]	[147]	[75]	[3]	[146]	[145]	[4]	[144]	[143]	[85]	[142]	[136]
Line model	π model	DPLM	DPLM ^a	DPLM ^a	π model	DPLM ^a	DPLM ^a	short line model	short line model	DPLM ^a	DPLM ^a	π model
Load model	static load	static load	static load	constant load	constant load	constant load	static load	constant load	constant load	static load	constant load	constant load
Load estimation	✓	✓	✓	–	–	–	✓	–	–	–	–	–
Non-homogeneity	✓	–	✓	✓	–	–	✓	–	✓	✓	–	✓
Unbalanced system	✓	–	✓	✓	–	–	✓	–	–	✓	–	✓
Laterals	✓	–	✓	✓	✓	–	✓	✓	✓	✓	✓	✓
Load taps	✓	–	✓	✓	–	✓	✓	✓	✓	✓	–	✓
Fault type	All	All	All	All	All	SLG ^d	SLG ^d	All	All	All	All	All
Section detection	–	–	CP ^b and PD ^c	–	–	Support vector machine loop	Voltage sag	–	–	–	–	–
Network Type	Radial	Radial	Radial	Radial	Radial	Radial	Radial	Radial	Radial	Radial	Radial	Radial and loop
Smart Grid	–	–	–	–	–	✓	–	✓	–	–	–	✓
Distributed generation	–	–	–	–	–	–	–	✓	–	–	–	✓
Time domain	–	✓	–	–	✓	–	–	–	–	–	–	–
Phase domain	✓	–	✓	✓	–	✓	✓	✓	✓	✓	✓	✓
Sequence domain	–	–	–	–	–	✓	✓	–	–	–	–	–

^a DPLM = distributed – parameter line model.

^b CP = current pattern.

^c PD = Protective devices.

^d SLG = Single-phase-to ground fault.

Table 3

Comparison of properties, advantages, and disadvantages in different methods of fault location in DNS.

References	Properties	Advantages	Disadvantages
[77]	–An impedance-based method –Two section estimation methods: First: based on the comparison between real data and simulated data of fault. Second: based on the comparison of those data frequency spectrum.	–It presents two section estimation methods –The section estimation methods are simpler compared to intelligent and state estimation-based methods –It only uses substation recorded data	–It is hard to generate rich enough simulated data for comparing with real recorded data –It is inapplicable for the networks with DGs –Needs accurate line parameters values –It needs the exact value of the load of each node –Needs rich data bank. –Its structure is complex. –Has high computational burden. –Needs special pulse generator devices. –Has slow time response. –Needs continues training. –Is inapplicable against changing in topology.
[145]	–Distributed line model is used. –Locates single-phase faults. –Distribution networks without DGs are considered. –A machine learning-based method is applied.	–Does not need to know the type of the fault. –No need to know the line parameters. –Estimates the real faulty section.	–Is inapplicable in face of loop networks. –Has low accuracy for unbalanced load. –Has low accuracy against high DGs penetration. –Has low reliability because of load characteristics and fault varieties. –Is inapplicable against changing in topology.
[4]	–Distributed line model is used. –Locates single-phase faults. –Distribution networks without DGs are considered. –Impedance-based method is applied.	–Has a simple structure. –Estimates real faulty section. –Needs only recorded data at the beginning of the feeder. –Applies phase domain data. –Has low computational burden.	–Is inapplicable in face of loop networks. –Has low accuracy for unbalanced load. –Has low accuracy against high DGs penetration. –Has low reliability because of load characteristics and fault varieties. –Is inapplicable against changing in topology.
[148]	– π line model is used. –Locates single-phase arc faults. –Distribution networks without DGs are considered. –Time domain equations-based method is applied.	–Does not need to know the type of the fault. –No need to know the line parameters. –Is applicable for loop networks.	–Needs high sampling rate devices. –Needs sampling devices in all buses of the network. –Highly sensitive to measurement error.
[136]	– π line model is used. –Locates all types of steady-state faults. –Distribution networks with DGs are considered. –Impedance-based method is applied.	–Has a simple structure. –Needs only recorded data at the beginning of the feeder. –Applies phase domain data. –Has low computational burden.	–Is inapplicable in face of loop networks. –Has low accuracy for unbalanced load. –Has low accuracy against high DGs penetration. –Has a low reliability because of load characteristics and fault varieties. –Is inapplicable against changing in topology. –Cannot detect real section of fault.
[149]			

Table 3 (continued)

References	Properties	Advantages	Disadvantages
	– π line model is used. –Locates high impedance arc faults. –Distribution networks without DGs are considered. –A searching-based method is applied.	–Does not need to know the type of the fault. –No need to know the line parameters. –Is applicable for loop networks.	–Needs high sampling rate devices. –Needs sampling devices in all buses of the network. –Highly sensitive to measurement error.
[150]	–Distributed line model is used. –Locates all types of steady-state faults. –Transmission lines are considered. –Time domain equations-based method is applied.	–Does not need to know the type of the fault. –Does not need to know the type of the fault. –No need to know the line parameters. –Is applicable for loop networks.	–Needs high sampling rate devices. –Needs sampling devices in all buses of the network. –Highly sensitive to measurement error.
[151]	–Distributed line model is used. –Locates all types of steady-state faults. –Distribution networks without DGs are considered. –Impedance-based method is applied.	–Has a simple structure. –Needs only recorded data at the beginning of the feeder. –Applies phase domain data. –Has low computational burden.	–Is inapplicable in face of loop networks. –Has low accuracy for the unbalanced load. –Has low accuracy against high DGs penetration. –Has low reliability because of load characteristics and fault varieties. –Cannot detect a real section of the fault.
[152]	–A state estimation-based method –Faulty zone separates with the rest of the network using revised state estimation method.	–Applicable on the networks with DGs –No need to locate measurement in all node –It can detect the location of fault in the network with high penetration DGs	–Needs accurate line parameters –Compared to Impedance-based methods it needs much more data –It has high computational burden compared to Impedance-based methods –The real faulty section can be estimated with higher accuracy
[137]	–An impedance-based method. –A section estimation method based on the least-squares error of the real and simulated voltage samples. Distributed line parameters are used.	–Low computational burden section estimation method –Applicable on both smart and traditional distribution networks –It only needs voltage and current of substation and DGs	–Needs accurate line parameters –The section estimation method needs an accurate simulation of fault which is almost impossible in the real word network because of uncertainties –The section estimation method is not robust against measurements error –It is inapplicable in face of high unbalance load flow and DGs penetration
[153]	–Distributed line model is used. –Locates all types of steady-state and transients' faults. –Distribution	–Does not need to know the type of the fault. –Does not need to know the type of the fault.	–Needs high sampling rate devices. –Needs sampling devices in all buses of the network.

(continued on next page)

Table 3 (continued)

References	Properties	Advantages	Disadvantages
	networks with all types of DGs are considered. –A hybrid time domain equations-based and meta-heuristic-based method is applied.	–No need to know the line parameters. –Is applicable for loop networks. –Line parameters are estimated using pre-fault data. –Is robust under different DGs operations and modes.	–Highly sensitive to measurement error.

proposed by institutions such as ETSI, IEEE, IEC, ISO and ANSI to ensure the smooth exchange of this information [154–156]. Table 4 shows some examples of these standards and their applications.

Ref. [157] presents a Wide-Area Traveling Wave Fault Location (WA-TWFL) technique that uses IEC61850. This algorithm initially identifies the network monitoring area where the disturbance source originates by examining the propagation times of the traveling wave via the extended double end technique. Subsequently, the line with the fault and the fault distance are found through chosen records from the detected disturbance area. The authors also provide data models for traveling waves derived from IEC61850, which supports interoperability and open communication between data acquisition devices and the WA-TWFL master station. The researchers in [158] presented a protection scheme based on traveling waves for an intelligent substation in distribution systems according to IEC61850. They have adopted WLAN communication based on the Sample Value (SV) and have calculated the SV packet size specifically for the traveling wave base fault location method. Nevertheless, they have not described the information modeling of traveling wave intelligent electronic devices for invoking IEC61850 SV services. Ref. [159] investigates the location of the fault in an IEEE 34 bus network via the SCADA system, current and voltage measurements, and the impedance-base technique. The communication of the data collection devices with the control equipment has been standardized using DNP3. These protocols play a crucial role in SCADA systems by providing communication between the SCADA master station and the smart electronic devices and Remote Terminal Units.

3.5.1. Fault location methods in smart grid

In [160], an impedance method is introduced for fault location in DNs, which can be used for single-source and multiple-source smart grids. Two algorithms have been presented for fault location in which synchronous voltages before fault and during fault received at multiple buses by PMUs, or data of the asynchronous AMI are used. Voltage drop

Table 4

Some examples of communication and automation standards in DNs and smart grid [154–156].

Name of Standard	Application/details
IEC 61850	Communication standard for the power utility automation
IEC 60870-5	Communication standard for telecontrol
IEC 60870-6	Communication between the control centers
IEC 60834	Teleprotection Equipment
IEC 61970	Energy management services application program interface (EMS-API)
IEC 61968	System interfaces for distribution management (Synchrophasor measurement)
IEC 61334	Distribution automation using distribution line carrier systems
IEC 61499	Distributed control and automation
IEEE C37.118.2	Synchrophasor Data Transfer
IEEE 1815 (DNP3)	Electric power systems communications- distributed network protocol
ETSI EN 300 132-3-1	Operating voltage range in normal condition for DC system
ANSI C12.22	Data network communications

measurements are used with a new impedance matrix for determining the location of various faults under noisy conditions. The proposed method does not require fault characteristics and impedance of the external network and the DGs. In [161], a fault location approach based on impedance is presented in smart distribution networks, which utilizes synchronized/non-synchronized types of measurements in the presence/absence of distributed generators. This approach only involves the measurement of voltage drops at a few buses and branch series impedance. Although drops in voltage at DG buses are to be assessed synchronously, synchronization amongst voltage drop measurements across the network is unessential.

In [162], a method is presented for determining the faulty sections in the smart grid, which is based on changing the old devices of the protection systems through intelligent switches with fault sensors that can transmit their state to the operational centers.

In [163], zero sequence transient current distribution after fault event is obtained at various frequencies using S-transform. The residual energy is used to describe the signal differences at the end of each section. The section with maximum residual energy is considered as the faulty section. This method does not require the voltage magnitude. The zero-sequence transient current after fault is considered as the analysis target. The residual energy between two ends of the faulty section is larger than two ends of a section without fault. This method is robust against fault location, fault angle, and transition resistance.

In [164], an adaptation method is presented based on load estimation and using PMU in DNs. The proposed algorithm does not require load distribution information and is not affected by load capacity, load type, and fault type. The self-adaptation measure after fault is calculated in real-time to prevent the effect of switching and DG output. This algorithm is suitable for monitoring short-circuit faults and determining their location in smart grids with high penetration DGs.

In [137], a novel impedance-based fault location algorithm has been presented for the smart distribution network equipped with micro-phasor measurement units, data loggers, and distributed generations. Two types of fully and not fully observable networks are considered. For the first type, the network is fully observable by a minimum number of micro-phasor measurement units. There are a limited number of micro-phasor measurement units in the network (on the substation and DGs as a minimum requirement) and data loggers on the rest nodes in the second type. A new two-stage algorithm is proposed to determine each node's initial and accurate load by applying the particle swarm optimization algorithm. As impedance-based algorithms determine several locations for fault in the network, a new simple least-squares error-based method, which has a lower computational burden than its other counterparts, has been presented.

3.5.2. Fault location methods in microgrid

Upon fault occurrence, the initial current and voltage traveling waves propagate in the DC microgrid. Therefore, fault location can be determined through analyzing characteristics of the traveling waves including time intervals at different locations [165], subsequent arrival times to a terminal, the time interval between the first arrivals at both terminals, and measuring the first arrival time of the traveling waves to the transformer stations [166]. However, fault location based on traveling waves requires efficient devices for data collection. Also, in microgrids with short distribution lines and complicated topology, many reflections occur that affect accuracy adversely. The arrival time of the waves is also short, which requires many samplings. Therefore, fault location based on traveling waves is not suitable for microgrids.

The methods based on active impedance estimation (AIE) and power probe unit (PPU) are used to determine fault location based on injection. In the fault location method based on AIE, after detecting a fault, a triangular waveform is injected by a power converter. Then, the impedance is calculated at the connection point. Finally, the fault is located by the reactive component [167]. In another method, a PPU is used to constitute the second-order RLC circuit through the fault route.

Then, the current response of the problem is analyzed, and fault distance is obtained in damped resonance frequency [168]. The essential advantage of these methods is that they do not require a communication system and high accuracy; but, their implementation cost is high since they require additional devices.

In [169], a fault location method is presented for single-phase microgrids. The fault is located by a linear relationship between the magnitude of the transient signal resulting from fault obtained by the sensor and the distance between the sensor and the fault location. As the distance between the sensor and the fault location increases, the magnitude of the received transient signals increases. Finally, an algorithm is designed to determine fault location based on a new relationship and integration of the information obtained from various sensors. The error of this method is $<10\%$.

Generally, to detect a fault and protect the network, a differential protection system is used. However, differential protection-based fault location methods have been investigated in various studies. Authors of [170] have measured the current at both ends of the line, and Ethernet cable (IEC 61850) is used to send the obtained information to both ends. In the first step, the fault is detected using the modified cumulative sum average method considering several differential current samples. A set of samples, including voltage and currents of both ends, are collected and used to calculate resistance and impedance from one end to the faulty point by the non-iterative Moore-Penrose pseudo-inverse technique. The main disadvantage of the differential methods depends on the presence of reliable and fast communication systems.

Authors of [171] have presented an online fault location method for islanded DC microgrids. The presented algorithm specified fault location based on transient states of current and voltage. Line-to-line and pole-to-ground faults are analyzed independently, and the algorithms are developed based on the current change rate. Transient measurements are recorded locally and used to locate short-circuit faults. While the communication-based method is used to locate impedance faults. The error of the proposed method for line-to-line short-circuits fault is 0.37% and 0.4% for pole-to-ground fault.

In [172], local current and voltage values and di/dt signals at each PD are measured for estimating the inductance between PD and fault. The online moving window least-squares method uses the samples at different times to estimate the inductance. The fault location accuracy can be increased using fast di/dt ; but making it noise-sensitive. Therefore, a digital and advanced method is required for calculating di/dt , and a digital filter is required to obtain the desired accuracy. Also, this method is based on a type of DC microgrid. In this microgrid, the capacitor is connected to one end of the cable; DC microgrids are not usually designed like this.

In [173], a method is proposed for locating faults in AC mesh microgrids. In there, a set of features measured from the signals are selected and extracted. Then, they are given to an SVM for fault detection. The presented method can determine fault locations for grid-connected and islanded microgrids with various configurations. A fault is usually detected in 20 ms, and the fault location is determined based on harmonic injection strategies in 80 ms.

In [174], a fault location method based on an impedance is presented for a three-phase microgrid. In this method, current and voltage measurements at both ends of the section are used to locate the fault. This method can be applied to various fault types. The maximum error of the presented method is obtained for resistance of $50\ \Omega$ and a line-to-line error of 0.8% .

In [175], a transient state of voltage and current is used to design an online method to locate the fault in the absence of communication systems in the islanded DC microgrids. A mathematical model is obtained for a faulty network which is used in the proposed algorithm for estimating the fault location. An estimation-based method is applied to this model to determine the accurate location and resistance of the fault. The internal faults with an error rate of $<2\%$ and reliability higher than 95% are located. The fault location time is 0.75 ms. The proposed

method can be used for both pole-to-pole and pole-to-ground faults. This algorithm does not depend on the operating point and the topology of the microgrid.

Another method for fault location based on the concept of the ratio of the transient voltage defined as the ratio of the measured voltages at both sides of the inductor terminal is defined in the time domain [176]. This method has two states: a single terminal and a double terminal. In the single terminal method, local measurement is used. However, if the fault resistance is large, its accuracy would be low. On the other hand, the double-terminal method's accuracy is higher, but it has an additional communication system and two voltage sensors.

In [177], a method is presented to detect and determine fault location in circular LVDC microgrids. This method is based on a multi-criterion system for fault detection and a neural network for fault location. This presented method requires a big data bank and is operated offline. Among the disadvantages of this method, information updating can be mentioned. The maximum error of the presented method is reported as 0.0089% .

Table 5 compares the fault location methods in microgrids.

3.6. A combined results and discussion for fault location methods in electrical distribution networks, smart grid, and microgrid

In this part, different fault location methods are investigated and reviewed for traditional and smart power distribution networks. According to the studies, it was observed that each of the methods is not able to determine the location of the fault 100% . Therefore, all cases and problems of the methods are based on the block diagram expressed in Figs. 25 and 31, so to overcome these problems, researchers try using hybrid methods, the data of various measuring instruments approved in the network with different sampling rate, pattern recognition algorithms and learning machine based on a large and accurate database, etc. furthermore, the accuracy of the presented methods is depended on some vital characteristics such as load model and data, fault resistances, line parameters, fault time, fault type, data resolution, updating data, data volume, network configuration, and instrument accuracy. Therefore, given the above, achieving a simple and comprehensive approach still requires deeper and broader research.

4. Future research

After reviewing and analyzing various methods for fault prediction and fault location in electrical distribution networks, future research directions are presented as follows:

- Electric vehicles are connected to the grid in two modes. In the first mode (G2V), the battery is charging and acting as a load. In the second mode (V2G), the battery is fully charged and can inject current into the grid. The effects of these two modes on fault location methods require further research.
- The current signal increases the measurement error due to its influence on the transformer saturation conditions. It is necessary to investigate a method for locating the fault that determines the fault location in electrical distribution networks using only the voltage at the beginning of the feeder.
- The key to the wide adoption of fault location algorithms is simplicity, cost-effectiveness, and low sampling rate data requirements. Using the least amount of data helps the simplicity and usability of the algorithm. A comprehensive fault location method using data only from the existing infrastructures such as smart meters needs further investigation.
- The presence of FACTS devices in electrical distribution networks compensates for the fault current and can cause an error in calculating the fault location. Therefore, it is necessary to provide a fault location method that is independent of the effects of FACTS devices.

Table 5
Comparison of fault location methods in microgrids.

Reference	[171]	[173]	[175]	[174]	[177]
Network Type	Loop	Loop	Loop	Radial	Loop
Fault type	LL & monopole ground	Three-phase fault to ground	LL&LG	All	LL&LG
Line model	Short Line Model	Short Line Model	Short Line Model	π Line Model	Short Line Model
Line Parameters	constant	constant	constant	constant	constant
Load estimation	–	–	–	✓	–
Non-homogeneity	✓	–	–	–	–
Unbalanced system	–	–	–	✓	–
Laterals	–	–	–	✓	–
Load taps	✓	✓	✓	✓	✓
Section detection	–	✓	–	✓	✓
Time domain	✓	–	✓	–	–
Phase domain	–	–	–	✓	–
Machin Learning	–	SVM	–	–	Neural Network
Island mode	✓	✓	✓	✓	✓
Grid connected mode	–	✓	–	✓	✓
Microgrid Type	DC	AC	DC	AC	DC

- The operation of microgrids in islanded or grid-connected modes may have an adverse effect on the fault location process. If a fault occurs in a microgrid when it is linked to the grid, its location is hardly detectable. This occurs because the microgrid is linked to a large and reliable grid, making fault location a difficult task. To address this issue, the impact of DGs controllers should be included in fault location formulation. In islanded mode, on the other hand, the level of fault current may vary owing to the DGs injected current, which alters the amount of fault current. Islanded grid fault location techniques should deal with this issue in future work to ensure appropriate functioning. Furthermore, for future study, techniques for locating faults should be resilient and adaptable, functioning correctly regardless of the network's operating mode.
- To anticipate faults, fault prediction methods often make use of a fundamental intelligent model, a limited collection of heuristic network faults, and a limited quantity of data. In future work, a mix of heuristic network data, meteorological data, and geospatial data can be used to provide a comprehensive platform for network failure prediction.
- Faults in network equipment, such as power transformers that need specialized chemicals to cool themselves, may also be the result of chemical or electrical problems. Utilizing chemical analysis and electrical data associated with equipment is another feasible alternative that has the potential to significantly improve the effectiveness of fault prediction methods that should be investigated in future research.
- Arcs inside the cable contribute to a portion of the fault in underground cables, which may be observed in the voltage or current signal. Along with heuristic fault data, spark patterns in underground cables coupled with historical records can be used to develop a more comprehensive and accurate fault prediction model.

5. Conclusion

Considering the importance of reliability in power systems, especially DNs, and the direct impact of fault, researchers have proposed various methods for fault prediction and fault location. In this paper, various modern methods proposed for fault prediction and location in DNs for increasing network reliability, fast recovery, optimal electric energy consumption, and customer satisfaction have been evaluated. In this paper, various faults are investigated and fault prediction methods have been studied. Then, the fault location in DN is divided into distance determination and detecting the faulty section due to the existence of various branches considering network devices and the used algorithm. Then, impedance methods, differential methods, and traveling waves have been studied for fault distance determination. Each method has advantages and disadvantages for implementation in the DN. In the

detection of the faulty section, implementation of a fault location algorithm is difficult and complicated since an accurate and massive data bank, high sampling rate and accurate information of the network are required. In some methods, a small change in the network topology changes all designs for fault location. Since the DNs have specific characteristics, various operation conditions and devices such as a capacitor, auto-booster, compensator, recloser, sectionalizer, cut-out fuse, and VIT switches for optimal operation, fault location in these networks would be complicated that requires a comprehensive approach that can respond all requirements of the DNs. With all of these interpretations, implementation problems for fault location and prediction are unresolved and require further research.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] J.M. Gers, E.J. Holmes, Protection of electricity distribution networks, IET (2004).
- [2] S. Zhang, Y. Wang, M. Liu, Z. Bao, Data-based line trip fault prediction in power systems using LSTM networks and SVM, IEEE Access 6 (2017) 7675–7686.
- [3] R. Dashti, J. Sadeh, Accuracy improvement of impedance-based fault location method for power distribution network using distributed-parameter line model, Int. Trans. Electr. Energy Syst. 24 (3) (2014) 318–334.
- [4] M. Daisy, R. Dashti, Single phase fault location in electrical distribution feeder using hybrid method, Energy 103 (2016) 356–368.
- [5] K. Chen, C. Huang, J. He, Fault detection, classification and location for transmission lines and distribution systems: a review on the methods, High Voltage 1 (1) (2016) 25–33.
- [6] A. Bahmanyar, S. Jamali, A. Estebasari, E. Bompard, A comparison framework for distribution system outage and fault location methods, Electr. Power Syst. Res. 145 (2017) 19–34.
- [7] M. Shafullah, M.A. Abido, A review on distribution grid fault location techniques, Electr. Power Compon. Syst. 45 (8) (2017) 807–824.
- [8] W. Javed, D. Chen, M.E. Farrag, Y. Xu, System configuration, fault detection, location, isolation and restoration: a review on LVDC microgrid protections, Energies 12 (6) (2019) 1001.
- [9] M.A. Aftab, S.S. Hussain, I. Ali, T.S. Ustun, Dynamic protection of power systems with high penetration of renewables: a review of the traveling wave based fault location techniques, Int. J. Electr. Power Energy Syst. 114 (2020) 105410.

- [10] A. Farughian, L. Kumpulainen, K. Kauhaniemi, Review of methodologies for earth fault indication and location in compensated and unearthed MV distribution networks, *Electr. Power Syst. Res.* 154 (2018) 373–380.
- [11] S. Gururajapathy, H. Mokhlis, H. Illias, Fault location and detection techniques in power distribution systems with distributed generation: a review, *Renew. Sustain. Energy Rev.* 74 (2017) 949–958.
- [12] N. Bayati, A. Hajizadeh, M. Soltani, Protection in DC microgrids: a comparative review, *IET Smart Grid* 1 (3) (2018) 66–75.
- [13] S. Beheshtaein, R. Cuzner, M. Savaghebi, J.M. Guerrero, Review on microgrids protection, *IET Gener. Transm. Distrib.* 13 (6) (2019) 743–759.
- [14] M. Monadi, M.A. Zamani, J.I. Candela, A. Luna, P. Rodriguez, Protection of AC and DC distribution systems Embedding distributed energy resources: a comparative review and analysis, *Renew. Sustain. Energy Rev.* 51 (2015) 1578–1593.
- [15] J. Fausling Olesen, H.R. Shaker, Predictive maintenance for pump systems and thermal power plants: state-of-the-art review, trends and challenges, *Sensors* 20 (8) (2020) 2425.
- [16] M.R. Skydt, M. Bang, H.R. Shaker, A probabilistic sequence classification approach for early fault prediction in distribution grids using long short-term memory neural networks, *Measurement* (2020) 108691.
- [17] R. Dashti, M. Daisy, M.H. Aliabadi, Healthy and faulty mode detection in power distribution networks based on park transformation, *Electr. Power Syst. Res.* 191 (2021).
- [18] R. Dashti, M. Daisy, S. Javadi, M.H. Aliabadi, Proposing a new method to improve the longitudinal differential relay performance using the Clarke transformation: theory, simulation, and experiment, *Measurement* 168 (2021).
- [19] H. Mohammadpour, R. Dashti, H.R. Shaker, A new practical approach for discrimination between inrush currents and internal faults in power transformers, *Technol. Econ. Smart Grids Sustainable Energy* 5 (1) (2020) 5.
- [20] M.T. Veljko, R.T. Predrag, M.D. Zeljko, Expert system for fault detection and isolation of coal-shortage in thermal power plants, in: 2010 Conference on Control and Fault-Tolerant Systems (SysTol), 2010, IEEE, pp. 666–671.
- [21] C. Nan, F. Khan, M. Iqbal, Abnormal process condition prediction (fault diagnosis) using G2 expert system, in: 2007 Canadian Conference on Electrical and Computer Engineering, IEEE, 2007, pp. 1507–1510.
- [22] Z. Yongli, H. Limin, L. Jinling, Bayesian networks-based approach for power systems fault diagnosis, *IEEE Trans. Power Delivery* 21 (2) (2006) 634–639.
- [23] Y. Brandt, B. Jervis, Y. Maidon, Circuit multi-fault diagnosis and prediction error estimation using a committee of Bayesian neural networks, 1997.
- [24] S.S. Rawat, V.A. Polavarapu, V. Kumar, E. Aruna, V. Sumathi, Anomaly detection in smart grid using rough set theory and K cross validation, in: 2014 International Conference on Circuits, Power and Computing Technologies [ICCPCT-2014], IEEE, 2014, pp. 479–483.
- [25] X. Xu, J.F. Peters, Rough set methods in power system fault classification, in: IEEE CCECE2002. Canadian conference on electrical and computer engineering. Conference proceedings (Cat. No. 02CH37373), vol. 1, IEEE, 2002, pp. 100–105.
- [26] Z. Jiang, Z. Li, N. Wu, M. Zhou, A Petri net approach to fault diagnosis and restoration for power transmission systems to avoid the output interruption of substations, *IEEE Syst. J.* 12 (3) (2017) 2566–2576.
- [27] S. Yang, A condition-based failure-prediction and processing-scheme for preventive maintenance, *IEEE Trans. Reliab.* 52 (3) (2003) 373–383.
- [28] W. Li, A. Monti, F. Ponci, Fault detection and classification in medium voltage DC shipboard power systems with wavelets and artificial neural networks, *IEEE Trans. Instrum. Meas.* 63 (11) (2014) 2651–2665.
- [29] S. Bhattacharya, Fault detection on a ring-main type power system network using artificial neural network and wavelet entropy method, in: International Conference on Computing, Communication & Automation, IEEE, 2015, pp. 1032–1037.
- [30] W. Guo, F. Wen, G. Ledwich, Z. Liao, X. He, J. Liang, An analytic model for fault diagnosis in power systems considering malfunctions of protective relays and circuit breakers, *IEEE Trans. Power Delivery* 25 (3) (2010) 1393–1401.
- [31] Y. Cui, J. Shi, Z. Wang, Power system fault reasoning and diagnosis based on the improved temporal constraint network, *IEEE Trans. Power Delivery* 31 (3) (2015) 946–954.
- [32] Y. Tao, J. Zheng, T. Wang, Y. Hu, A state and fault prediction method based on RBF neural networks, in: 2016 IEEE Workshop on Advanced Robotics and its Social Impacts (ARSO), IEEE, 2016, pp. 221–225.
- [33] C. Xu, G. Wang, X. Liu, D. Guo, T.-Y. Liu, Health status assessment and failure prediction for hard drives with recurrent neural networks, *IEEE Trans. Comput.* 65 (11) (2016) 3502–3508.
- [34] T. De Bruin, K. Verbert, R. Babuška, Railway track circuit fault diagnosis using recurrent neural networks, *IEEE Trans. Neural Netw. Learn. Syst.* 28 (3) (2016) 523–533.
- [35] S.U. Jan, Y.-D. Lee, J. Shin, I. Koo, Sensor fault classification based on support vector machine and statistical time-domain features, *IEEE Access* 5 (2017) 8682–8690.
- [36] W.-S. Huang, X. Lu, Y. Liu, Q. Chen, M.-H. Qi, and H.-J. Gao, Fault prediction of distribution network based on support vector machine, *DEStech Transactions on Engineering and Technology Research*, no. amee, 2019.
- [37] P.Q.D.A.W. Group, Electric Signatures of Power Equipment Failures: IEEE PES-TR73, in: IEEE Power & Energy Society, Technical Report, 2019.
- [38] Distribution Fault Location and Waveform Characterization, EPRI, Palo Alto, CA, 2009, pp. 1017842.
- [39] Y. Bai, Y. Li, Y. Liu, Z. Ma, Short-term prediction of distribution network faults based on support vector machine, in: 2017 12th IEEE Conference on Industrial Electronics and Applications (ICIEA), IEEE, 2017, pp. 1421–1426.
- [40] T. Gu, J. Janssen, E. Tazelaar, G. Popma, Risk prediction in distribution networks based on the relation between weather and (underground) component failure, *CIREN-Open Access Proc. J.* 2017 (1) (2017) 1442–1445.
- [41] Y. Diao, R. Huang, C. Wang, D. Jia, Fault risk prevention model of distribution network based on Hidden Markov, in: 2018 5th International Conference on Information Science and Control Engineering (ICISCE), IEEE, 2018, pp. 1123–1127.
- [42] G. Zheng, G. Radman, W. Guan, S. Yang, Wide area phasor measurements based disturbance monitoring for line trip event, in: 2013 IEEE Power & Energy Society General Meeting, IEEE, 2013, pp. 1–5.
- [43] D. Zhou, Y. Liu, J. Dong, Frequency-based real-time line trip detection and alarm trigger development, in: 2014 IEEE PES General Meeting| Conference & Exposition, IEEE, 2014, pp. 1–5.
- [44] M.R. Skydt, M. Bang, H.R. Shaker, A probabilistic sequence classification approach for early fault prediction in distribution grids using long short-term memory neural networks, *Measurement* 170 (2021) 108691.
- [45] L. Fan, J. Li, Y. Pan, S. Wang, C. Yan, D. Yao, Research and application of smart grid early warning decision platform based on big data analysis, in: 2019 4th International Conference on Intelligent Green Building and Smart Grid (IGBSG), IEEE, 2019, pp. 645–648.
- [46] M. Xiang, J. Min, Z. Wang, P. Gao, A novel fault early warning model based on fault gene table for smart distribution grids, *Energies* 10 (12) (2017) 1963.
- [47] C. Haseltine, E. E.-S. Eman, Prediction of power grid failure using neural network learning, in: 2017 16th IEEE International Conference on Machine Learning and Applications (ICMLA), IEEE, 2017, pp. 505–510.
- [48] I. Standard, Testing and Measurement Techniques—Power Quality Measurement Methods, 2003.
- [49] G. Olguin, M. Aedo, M. Arias, A. Ortiz, A Monte Carlo simulation approach to the method of fault positions for stochastic assessment of voltage dips (sags), in: 2005 IEEE/PES Transmission & Distribution Conference & Exposition: Asia and Pacific, IEEE, 2005, pp. 1–6.
- [50] M. Moschakis, N. Hatziaargyriou, Analytical calculation and stochastic assessment of voltage sags, *IEEE Trans. Power Delivery* 21 (3) (2006) 1727–1734.
- [51] C.-H. Park, G. Jang, Stochastic estimation of voltage sags in a large meshed network, *IEEE Trans. Power Delivery* 22 (3) (2007) 1655–1664.
- [52] J.A. Martinez, J. Martin-Arnedo, Voltage sag stochastic prediction using an electromagnetic transients program, *IEEE Trans. Power Delivery* 19 (4) (2004) 1975–1982.
- [53] Y. Zhang, J. Milanovic, Techno-economic assessment of voltage sag performance and mitigation, *Citeseer* (2008).
- [54] A. dos Santos, M.T.C. de Barros, Predicting equipment outages due to voltage sags, *IEEE Trans. Power Delivery* 31 (4) (2015) 1683–1691.
- [55] G. Hussain, Methods for Arc-flash Prediction in MV/LV Switchgear, Aalto University, 2011. M. Sc. Thesis.
- [56] J.A. Kay, J. Arvola, L. Kumpulainen, Protecting at the speed of light, *IEEE Ind. Appl. Mag.* 17 (3) (2011) 12–18.
- [57] L. Kumpulainen, G.A. Hussain, M. Lehtonen, J.A. Kay, Preemptive arc fault detection techniques in switchgear and controlgear, *IEEE Trans. Ind. Appl.* 49 (4) (2013) 1911–1919.
- [58] M.A. Stuchly, S.S. Stuchly, Coaxial line reflection methods for measuring dielectric properties of biological substances at radio and microwave frequencies – a review, *IEEE Trans. Instrum. Meas.* 29 (3) (1980) 176–183.
- [59] G.B. Gajda, S.S. Stuchly, Numerical analysis of open-ended coaxial lines, *IEEE Trans. Microw. Theory Tech.* 31 (5) (1983) 380–384.
- [60] I.A. Metwally, D-dot probe for fast-front high-voltage measurement, *IEEE Trans. Instrum. Meas.* 59 (8) (2009) 2211–2219.
- [61] G.A. Hussain, M. Shafiq, M. Lehtonen, M. Hashmi, Online condition monitoring of MV switchgear using D-dot sensor to predict arc-faults, *IEEE Sens. J.* 15 (12) (2015) 7262–7272.
- [62] B.D. Russell, B.M. Aucoin, Energy analysis fault detection system, ed: Google Patents, 1996.
- [63] K. Muthu-Manivannan, C.L. Benner, P. Xu, B.D. Russell, Arcing event detection, ed: Google Patents, 2011.
- [64] A. Sundaram, Distribution fault location: field data and analysis, EPRI (1012438), 2006.
- [65] C.L. Benner, B.D. Russell, Practical high-impedance fault detection on distribution feeders, *IEEE Trans. Ind. Appl.* 33 (3) (1997) 635–640.
- [66] B.D. Russell, Expert system for detecting high impedance faults, ed: Google Patents, 1996.
- [67] F. Gatta, A. Geri, S. Lauria, M. Maccioni, Analytical prediction of abnormal temporary overvoltages due to ground faults in MV networks, *Electr. Power Syst. Res.* 77 (10) (2007) 1305–1313.
- [68] Y. Tang, F. Li, Q. Wang, Y. Xu, Hybrid method for power system transient stability prediction based on two-stage computing resources, *IET Gener. Transm. Distrib.* 12 (8) (2017) 1697–1703.
- [69] S.F. Stefenon, et al., Wavelet group method of data handling for fault prediction in electrical power insulators, *Int. J. Electr. Power Energy Syst.* 123 (2020) 106269.
- [70] Distribution Fault Anticipation: Phase III: System Integration and Library Enhancement, EPRI, Palo Alto, CA, 2009, pp. 1016036.
- [71] G. Hinton, et al., Deep neural networks for acoustic modeling in speech recognition: the shared views of four research groups, *IEEE Signal Process Mag.* 29 (6) (2012) 82–97.
- [72] B. Jian, B.C. Vemuri, Robust point set registration using gaussian mixture models, *IEEE Trans. Pattern Anal. Mach. Intell.* 33 (8) (2010) 1633–1645.

- [73] J. Jiang, R. Chen, M. Chen, W. Wang, C. Zhang, Dynamic fault prediction of power transformers based on hidden Markov model of dissolved gases analysis, *IEEE Trans. Power Delivery* 34 (4) (2019) 1393–1400.
- [74] P. Rai, N.D. Londhe, R. Raj, Fault classification in power system distribution network integrated with distributed generators using CNN, *Electr. Power Syst. Res.* (2020) 106914.
- [75] R. Dashti, J. Sadeh, Applying dynamic load estimation and distributed-parameter line model to enhance the accuracy of impedance-based fault-location methods for power distribution networks, *Electr. Power Compon. Syst.* 41 (14) (2013) 1334–1362.
- [76] S.-J. Lee, et al., An intelligent and efficient fault location and diagnosis scheme for radial distribution systems, *IEEE Trans. Power Delivery* 19 (2) (2004) 524–532.
- [77] R. Dashti, J. Sadeh, Fault section estimation in power distribution network using impedance-based fault distance calculation and frequency spectrum analysis, *IET Gener. Transm. Distrib.* 8 (8) (2014) 1406–1417.
- [78] J. Mora-Florez, J. Melendez, G. Carrillo-Calcado, Comparison of impedance based fault location methods for power distribution systems, *Electr. Power Syst. Res.* 78 (4) (2008) 657–666.
- [79] R.H. Salim, M. Resener, A.D. Filomena, K.R.C. De Oliveira, A.S. Bretas, Extended fault-location formulation for power distribution systems, *IEEE Trans. Power Delivery* 24 (2) (2009) 508–516.
- [80] A.D. Filomena, M. Resener, R.H. Salim, A.S. Bretas, Fault location for underground distribution feeders: an extended impedance-based formulation with capacitive current compensation, *Int. J. Electr. Power Energy Syst.* 31 (9) (2009) 489–496.
- [81] Distribution Fault Location: Circuit Model Errors, Current-Only Fault Location, and Momentary Fault Events, EPRI, Palo Alto, CA, 2010, pp. 1020094.
- [82] Benchmarking of Fault-Location Technologies, EPRI, Palo Alto, CA, 2011, pp. 1022730.
- [83] R. Salim, K. Salim, A. Bretas, Further improvements on impedance-based fault location for power distribution systems, *IET Gener. Transm. Distrib.* 5 (4) (2011) 467–478.
- [84] H. Nouri, M.M. Alamuti, Comprehensive distribution network fault location using the distributed parameter model, *IEEE Trans. Power Delivery* 26 (4) (2011) 2154–2162.
- [85] R. Dashti, M. Daisy, H.R. Shaker, M. Tahavori, Impedance-based fault location method for four-wire power distribution networks, *IEEE Access* 6 (2017) 1342–1349.
- [86] K. Kalita, S. Anand, S. Parida, A novel non-iterative fault location algorithm for transmission line with unsynchronized terminal, *IEEE Trans. Power Delivery* (2021).
- [87] B. Lian, M. Salama, An overview of digital fault location algorithms for power transmission lines using transient waveforms, *Electr. Power Syst. Res.* 29 (1) (1994) 17–25.
- [88] A. Borghetti, M. Bosetti, C. Nucci, M. Paolone, A. Abur, Integrated use of time-frequency wavelet decompositions for fault location in distribution networks: theory and experimental validation, *IEEE Trans. Power Delivery* 25 (4) (2010) 3139–3146.
- [89] A. Fathy, R. Dashti, M. Najafi, H.R. Shaker, Transient and steady-state faults location in intelligent distribution networks compensated with D-STATCOM using time-domain equations and distributed line model, *Electr. Eng.* (2021) 1–16.
- [90] M. Daisy, R. Dashti, H.R. Shaker, A new fault-location method for HVDC transmission-line based on DC components of voltage and current under line parameter uncertainty, *Electr. Eng.* 99 (2) (2017) 573–582.
- [91] D. Jalali, N. Moslemi, Fault location for radial distribution systems using fault generated high-frequency transients and wavelet analysis, in: *CIREP 2005-18th International Conference and Exhibition on Electricity Distribution*, IET, 2005, pp. 1–4.
- [92] A.M. Elhaffar, Power transmission line fault location based on current traveling waves, 2008.
- [93] K.J. Ferreira, Fault Location for Power Transmission Systems Using Magnetic Field Sensing Coils, 2007.
- [94] K.C. De Oliveira, R.H. Salim, A. Shuck Jr, A.S. Bretas, Faulted branch identification on power distribution systems under noisy environment. *International Conference on Power Systems Transients*, 2009.
- [95] O. Naidu, A.K. Pradhan, A traveling wave-based fault location method using unsynchronized current measurements, *IEEE Trans. Power Delivery* 34 (2) (2018) 505–513.
- [96] M. Vopálenský, A. Platil, P. Kašpar, Wattmeter with AMR sensor, *Sens. Actuators, A* 123 (2005) 303–307.
- [97] C. Reig, M.-D. Cubells-Beltrán, D. Ramírez Muñoz, Magnetic field sensors based on giant magnetoresistance (GMR) technology: applications in electrical current sensing, *Sensors* 9 (10) (2009) 7919–7942.
- [98] K. Zhu, W.K. Lee, P.W. Pong, Non-contact capacitive-coupling-based and magnetic-field-sensing-assisted technique for monitoring voltage of overhead power transmission lines, *IEEE Sens. J.* 17 (4) (2016) 1069–1083.
- [99] C. Zheng, et al., Magnetoresistive sensor development roadmap (non-recording applications), *IEEE Trans. Magn.* 55 (4) (2019) 1–30.
- [100] A.H. Khawaja, Q. Huang, J. Li, Z. Zhang, Estimation of current and sag in overhead power transmission lines with optimized magnetic field sensor array placement, *IEEE Trans. Magn.* 53 (5) (2017) 1–10.
- [101] Y. Chen, Q. Huang, A.H. Khawaja, Interference-rejecting current measurement method with tunnel magnetoresistive magnetic sensor array, *IET Sci. Meas. Technol.* 12 (6) (2018) 733–738.
- [102] J. Neto, C. Sartori, E. Blanco, J. Huillery, Auto-regressive model of magnetic signature for non-invasive fault location in power systems, 2015.
- [103] N. Bahador, F. Namdari, H.R. Matinfar, Tree-related high impedance fault location using phase shift measurement of high frequency magnetic field, *Int. J. Electr. Power Energy Syst.* 100 (2018) 531–539.
- [104] N.R. Bahador, F. Namdari, H.R. Matinfar, Feature extraction of tree-related high impedance faults as a source of electromagnetic interference around medium voltage power lines' corridors, *Prog. Electromag. Res.* 75 (2017) 13–26.
- [105] M. Kazim, A.H. Khawaja, U. Zabit, Q. Huang, Fault detection and localization for overhead 11-kV distribution lines with magnetic measurements, *IEEE Trans. Instrum. Meas.* 69 (5) (2019) 2028–2038.
- [106] A. Tashakkori, P.J. Wolfs, S. Islam, A. Abu-Siada, Fault location on radial distribution networks via distributed synchronized traveling wave detectors, *IEEE Trans. Power Delivery* 35 (3) (2019) 1553–1562.
- [107] P. Bountouris, H. Guo, D. Tzelepis, I. Abdulhadi, F. Coffe, C. Booth, MV faulted section location in distribution systems based on unsynchronized LV measurements, *Int. J. Electr. Power Energy Syst.* 119 (2020) 105882.
- [108] J.A.D.O. Neto, C.A.F. Sartori, G.M. Junior, Fault location in overhead transmission lines based on magnetic signatures and on the extended kalman filter, *IEEE Access* 9 (2021) 15259–15270.
- [109] R. Das, M. Sachdev, T. Sidhu, A fault locator for radial subtransmission and distribution lines, in: *2000 Power Engineering Society Summer Meeting (Cat. No. 00CH37134)*, vol. 1, IEEE, 2000, pp. 443–448.
- [110] R. Dashti, J. Sadeh, Fault indicator allocation in power distribution network for improving reliability and fault section estimation, in: *2011 International Conference on Advanced Power System Automation and Protection*, vol. 2, IEEE, 2011, pp. 1406–1411.
- [111] Z. Wu, I. Voloh, H. Lihan, Z. Xu, Fault location system and method for distribution network, ed: Google Patents, 2016.
- [112] R. Dashti, J. Sadeh, Fault section estimation in power distribution network only with special cutout fuse links setting, *Woulfenia J.* (2012) 317–323.
- [113] R. Dashti, J. Sadeh, A new method for fault section estimation in distribution network, in: *2010 International Conference on Power System Technology*, IEEE, 2010, pp. 1–5.
- [114] R. Dashti, J. Sadeh, A new practical method for fault section estimation in distribution system, *J. Iran. Assoc. Electr. Electron. Eng.* 10 (1) (2013) 47–63.
- [115] M. Gholami, A. Abbaspour, M. Moeini-Aghaie, M. Fotuhi-Firuzabad, M. Lehtonen, Detecting the location of short-circuit faults in active distribution network using PMU based state estimation, *IEEE Trans. Smart Grid* (2019).
- [116] M. Gholami, A.A.T. Fard, M. Moeini-Aghaie, Linear voltage based state estimator for active distribution system including phasor measurement unit (PMU), in: *2018 Electrical Power Distribution Conference (EPDC)*, 2018, IEEE, pp. 1–6.
- [117] M. Pignati, L. Zanni, P. Romano, R. Cherkaoui, M. Paolone, Fault detection and faulted line identification in active distribution networks using synchrophasors-based real-time state estimation, *IEEE Trans. Power Delivery* 32 (1) (2016) 381–392.
- [118] A. Primadianto, C.-N. Lu, A review on distribution system state estimation, *IEEE Trans. Power Syst.* 32 (5) (2016) 3875–3883.
- [119] S.A. Hosseini, J. Sadeh, B. Mozafari, Robust wide-area impedance-based fault location method utilising LAV estimator, *IET Gener. Transm. Distrib.* 10 (10) (2016) 2475–2485.
- [120] S. Jamali, A. Bahmanyar, E. Bompard, Fault location method for distribution networks using smart meters, *Measurement* 102 (2017) 150–157.
- [121] M.U. Usman, M.O. Faruque, Validation of a PMU-based fault location identification method for smart distribution network with photovoltaics using real-time data, *IET Gener. Transm. Distrib.* 12 (21) (2018) 5824–5833.
- [122] S. Barman, B.K.S. Roy, Detection and location of faults in large transmission networks using minimum number of phasor measurement units, *IET Gener. Transm. Distrib.* 12 (8) (2018) 1941–1950.
- [123] R. Dashti, M. Tahavori, M. Daisy, H.R. Shaker, A new matching algorithm for fault section estimation in power distribution networks, in: *2018 International Symposium on Advanced Electrical and Communication Technologies (ISAECT)*, 2018, IEEE, pp. 1–4.
- [124] R. Dashti, M. Daisy, S. Javadi, A new simple matching algorithm presentation for fault section estimation in power distribution network, *Iran. Electr. Ind. J. Quality Prod.* 7 (14 autumn and winter) (2019).
- [125] E. Gord, R. Dashti, M. Najafi, H.R. Shaker, Real fault section estimation in electrical distribution networks based on the fault frequency component analysis, *Energies* 12 (6) (2019) 1145.
- [126] K.R.C. de Oliveira, R.H. Salim, A.D. Filomena, M. Resener, A.S. Bretas, Unbalanced underground distribution systems fault detection and section estimation, in: *International Conference on Intelligent Computing*, Springer, 2007, pp. 1054–1065.
- [127] R.A.F. Pereira, L.G.W. da Silva, M. Kezunovic, J.R.S. Mantovani, Improved fault location on distribution feeders based on matching during-fault voltage sags, *IEEE Trans. Power Delivery* 24 (2) (2009) 852–862.
- [128] J. Mora-Florez, G. Morales-Espana, S. Pérez-Londoño, Learning-based strategy for reducing the multiple estimation problem of fault zone location in radial power systems, *IET Gener. Transm. Distrib.* 3 (4) (2009) 346–356.
- [129] M. Pourahmadi-Nakhli, A.A. Safavi, Path characteristic frequency-based fault locating in radial distribution systems using wavelets and neural networks, *IEEE Trans. Power Delivery* 26 (2) (2010) 772–781.
- [130] H. Mokhlis, H. Li, Non-linear representation of voltage sag profiles for fault location in distribution networks, *Int. J. Electr. Power Energy Syst.* 33 (1) (2011) 124–130.
- [131] R. Dashti, M. Daisy, H.R. Shaker, A new method presentation for locating fault in power distribution networks, in: *2016 19th International Symposium on Electrical Apparatus and Technologies (SIELA)*, IEEE, 2016, pp. 1–4.

- [132] H. Mokhlis, A. Khalid, H. Li, Voltage sags pattern recognition technique for fault section identification in distribution networks, in: 2009 IEEE Bucharest PowerTech, IEEE, 2009, pp. 1–6.
- [133] J. Sadeh, E. Bakhshizadeh, R. Kazemzadeh, A new fault location algorithm for radial distribution systems using modal analysis, *Int. J. Electr. Power Energy Syst.* 45 (1) (2013) 271–278.
- [134] A. Rafinia, J. Moshtagh, A new approach to fault location in three-phase underground distribution system using combination of wavelet analysis with ANN and FLS, *Int. J. Electr. Power Energy Syst.* 55 (2014) 261–274.
- [135] A.C. Adewole, R. Tzoneva, S. Behardien, Distribution network fault section identification and fault location using wavelet entropy and neural networks, *Appl. Soft Comput.* 46 (2016) 296–306.
- [136] R. Dashti, M. Ghasemi, M. Daisy, Fault location in power distribution network with presence of distributed generation resources using impedance based method and applying π line model, *Energy* 159 (2018) 344–360.
- [137] H. Mirshekari, R. Dashti, A. Keshavarz, A.J. Torabi, H.R. Shaker, A novel fault location methodology for smart distribution networks, *IEEE Trans. Smart Grid* (2020).
- [138] K. Jia, T. Bi, Z. Ren, D.W. Thomas, M. Sumner, High frequency impedance based fault location in distribution system with DGs, *IEEE Trans. Smart Grid* 9 (2) (2016) 807–816.
- [139] A. Bahmanyar, S. Jamali, Fault location in active distribution networks using non-synchronized measurements, *Int. J. Electr. Power Energy Syst.* 93 (2017) 451–458.
- [140] X. Wang, et al., Location of single phase to ground faults in distribution networks based on synchronous transients energy analysis, *IEEE Trans. Smart Grid* 11 (1) (2019) 774–785.
- [141] T. Zhang, H. Yu, P. Zeng, L. Sun, C. Song, J. Liu, Single phase fault diagnosis and location in active distribution network using synchronized voltage measurement, *Int. J. Electr. Power Energy Syst.* 117 (2020) 105572.
- [142] R.A. de Aguiar, A.L. Dalcastagne, H.H. Zürn, R. Seara, Impedance-based fault location methods: sensitivity analysis and performance improvement, *Electr. Power Syst. Res.* 155 (2018) 236–245.
- [143] M.A. Gabr, D.K. Ibrahim, E.S. Ahmed, M.I. Gilany, A new impedance-based fault location scheme for overhead unbalanced radial distribution networks, *Electr. Power Syst. Res.* 142 (2017) 153–162.
- [144] R. Chen, T. Lin, R. Bi, X. Xu, Novel strategy for accurate locating of voltage sag sources in smart distribution networks with inverter-interfaced distributed generators, *Energies* 10 (11) (2017) 1885.
- [145] X. Deng, R. Yuan, Z. Xiao, T. Li, K.L.L. Wang, Fault location in loop distribution network using SVM technology, *Int. J. Electr. Power Energy Syst.* 65 (2015) 254–261.
- [146] R. Liang, G. Fu, X. Zhu, X. Xue, Fault location based on single terminal travelling wave analysis in radial distribution network, *Int. J. Electr. Power Energy Syst.* 66 (2015) 160–165.
- [147] M.M. Alamuti, H. Nouri, R.M. Ciric, V. Terzija, Intermittent fault location in distribution feeders, *IEEE Trans. Power Delivery* 27 (1) (2011) 96–103.
- [148] A. Herrera-Orozco, A. Bretas, C. Orozco-Henao, L. Iurinic, J. Mora-Flórez, Incipient fault location formulation: a time-domain system model and parameter estimation approach, *Int. J. Electr. Power Energy Syst.* 90 (2017) 112–123.
- [149] S.H. Mortazavi, Z. Moravej, S.M. Shahrtash, A searching based method for locating high impedance arcing fault in distribution networks, *IEEE Trans. Power Delivery* 34 (2) (2018) 438–447.
- [150] C. Wang, Z. Yun, Parameter-free fault location algorithm for distribution network T-type transmission lines, *Energies* 12 (8) (2019) 1534.
- [151] F. Aboshady, D. Thomas, M. Sumner, A new single end wideband impedance based fault location scheme for distribution systems, *Electr. Power Syst. Res.* 173 (2019) 263–270.
- [152] M. Gholami, A. Abbaspour, M. Moeini-Aghaie, M. Fotuhi-Firuzabad, M. Lehtonen, Detecting the location of short-circuit faults in active distribution network using PMU-based state estimation, *IEEE Trans. Smart Grid* 11 (2) (2019) 1396–1406.
- [153] H. Mirshekari, R. Dashti, H.R. Shaker, R. Samsami, A.J. Torabi, Linear and nonlinear fault location in smart distribution network under line parameter uncertainty, *IEEE Trans. Ind. Inf.* (2021).
- [154] A. Naumann, I. Bielchev, N. Voropai, Z. Styczynski, Smart grid automation using IEC 61850 and CIM standards, *Control Eng. Pract.* 25 (2014) 102–111.
- [155] E. Kabalci, Y. Kabalci, *Smart Grids and Their Communication Systems*, Springer, 2019.
- [156] V.C. Gungor, et al., Smart grid technologies: communication technologies and standards, *IEEE Trans. Ind. Inf.* 7 (4) (2011) 529–539.
- [157] Y. Chen, D. Liu, B. Xu, Wide-area traveling wave fault location system based on IEC61850, *IEEE Trans. Smart Grid* 4 (2) (2012) 1207–1215.
- [158] N.H. Ali, B.M. Ali, O. Basir, M.L. Othman, F. Hashim, K. Abdel-Latif, Protection of smart substation based on WLAN complies with IEC 61850 using traveling wave analysis, *Electr. Power Syst. Res.* 140 (2016) 20–26.
- [159] A. Rangel-Damián, E. Melgoza-Vázquez, H. F. Ruíz-Paredes, Application of fault location methods in distribution circuits with SCADA, in: 2017 IEEE International Autumn Meeting on Power, Electronics and Computing (ROPEC), IEEE, 2017, pp. 1–6.
- [160] M. Majidi, M. Etezadi-Amoli, A new fault location technique in smart distribution networks using synchronized/nonsynchronized measurements, *IEEE Trans. Power Delivery* 33 (3) (2017) 1358–1368.
- [161] M. Majidi, M. Etezadi-Amoli, A new fault location technique in smart distribution networks using synchronized/nonsynchronized measurements, *IEEE Trans. Power Delivery* 33 (3) (2018) 1358–1368.
- [162] P. Campos, R. Buzo, F. Leão, Fault section location in smart grid using smart switches, in: 2018 Simposio Brasileiro de Sistemas Eletricos (SBSE), IEEE, 2018, pp. 1–6.
- [163] Z. Pang et al., A fault section location method based on energy remainder of generalized S-transform for single-phase ground fault of distribution networks, in: 2018 IEEE 3rd Advanced Information Technology, Electronic and Automation Control Conference (IAEAC), IEEE, 2018, pp. 1511–1515.
- [164] X. Xing et al., Short-circuit Fault Location method for Smart Distribution Network Considering load estimation, in: 2018 2nd IEEE Conference on Energy Internet and Energy System Integration (EI2), IEEE, 2018, pp. 1–5.
- [165] O.K. Nanayakkara, A.D. Rajapakse, R. Wachal, Location of DC line faults in conventional HVDC systems with segments of cables and overhead lines using terminal measurements, *IEEE Trans. Power Delivery* 27 (1) (2012) 279–288.
- [166] S. Azizi, M. Sanaye-Pasand, M. Abedini, A. Hasani, A traveling-wave-based methodology for wide-area fault location in multiterminal DC systems, *IEEE Trans. Power Delivery* 29 (6) (2014) 2552–2560.
- [167] K. Jia, T. Bi, B. Liu, E. Christopher, D.W. Thomas, M. Sumner, Marine power distribution system fault location using a portable injection unit, *IEEE Trans. Power Delivery* 30 (2) (2014) 818–826.
- [168] R. Mohanty, U.S.M. Balaji, A.K. Pradhan, An accurate noniterative fault-location technique for low-voltage DC microgrid, *IEEE Trans. Power Delivery* 31 (2) (2015) 475–481.
- [169] J. Duan, K. Zhang, L. Cheng, A novel method of fault location for single-phase microgrids, *IEEE Trans. Smart Grid* 7 (2) (2015) 915–925.
- [170] S. Dhar, R.K. Patnaik, P. Dash, Fault detection and location of photovoltaic based DC microgrid using differential protection strategy, *IEEE Trans. Smart Grid* 9 (5) (2017) 4303–4312.
- [171] A. Meghwani, S. Chakrabarti, S. Srivastava, An on-line fault location technique for DC microgrid using transient measurements, in: 2017 7th International Conference on Power Systems (ICPS), IEEE, 2017, pp. 386–391.
- [172] X. Feng, L. Qi, J. Pan, A novel fault location method and algorithm for DC distribution protection, *IEEE Trans. Ind. Appl.* 53 (3) (2017) 1834–1840.
- [173] S. Beheshtaein, R. Cuzner, M. Savaghebi, S. Golestan, J.M. Guerrero, Fault location in microgrids: a communication-based high-frequency impedance approach, *IET Gener. Transm. Distrib.* 13 (8) (2018) 1229–1237.
- [174] R. Dutta, S. Samantaray, Assessment of impedance based fault locator for AC micro-grid, *Renew. Energy Focus* 26 (2018) 1–10.
- [175] A. Meghwani, S.C. Srivastava, S. Chakrabarti, Local measurement-based technique for estimating fault location in multi-source DC microgrids, *IET Gener. Transm. Distrib.* 12 (13) (2018) 3305–3313.
- [176] S. Jiang, C. Fan, N. Huang, Y. Zhu, M. He, A Fault Location Method for DC Lines Connected With DAB Terminal in Power Electronic Transformer, *IEEE Trans. Power Delivery* 34 (1) (2018) 301–311.
- [177] A. Abdali, K. Mazlumi, R. Noroozian, High-speed fault detection and location in DC microgrids systems using Multi-Criterion System and neural network, *Appl. Soft Comput.* 79 (2019) 341–353.